U-bend evaluation: measured tables

dataset summary

Table 1: Successful U-bend design data in released coordinate/physical target units. Inputs are signed geometry parameters, pressure is Pa, and the cooling objective is K squared times m squared. No error-based trimming or global standardization is applied.

Variable	count	mean	std	min	25%	50%	75%	max
input_0	8946.000	0.001	0.578	-1.000	-0.499	0.001	0.501	1.000
input_1	8946.000	-0.008	0.576	-1.000	-0.505	-0.011	0.489	1.000
input_2	8946.000	-0.001	0.577	-1.000	-0.499	-0.001	0.498	1.000
input_3	8946.000	0.000	0.577	-1.000	-0.499	-0.002	0.499	1.000
input_4	8946.000	0.004	0.576	-1.000	-0.493	0.004	0.503	1.000
input_5	8946.000	-0.005	0.576	-1.000	-0.503	-0.007	0.493	1.000
input_6	8946.000	0.505	0.286	0.000	0.259	0.505	0.751	1.000
input_7	8946.000	0.023	0.554	-1.000	-0.452	0.025	0.500	1.000
input_8	8946.000	-0.001	0.574	-1.000	-0.496	-0.003	0.493	1.000
input_9	8946.000	0.517	0.286	0.000	0.273	0.524	0.765	1.000
input_10	8946.000	0.506	0.286	0.000	0.260	0.508	0.754	1.000
input_11	8946.000	0.029	0.552	-1.000	-0.441	0.031	0.503	1.000
input_12	8946.000	0.500	0.289	0.000	0.250	0.500	0.750	1.000
input_13	8946.000	0.500	0.289	0.000	0.249	0.500	0.748	1.000
input_14	8946.000	0.500	0.289	0.000	0.250	0.499	0.750	1.000
input_15	8946.000	0.501	0.289	0.000	0.251	0.501	0.751	1.000
input_16	8946.000	0.500	0.289	0.000	0.249	0.500	0.751	1.000
input_17	8946.000	0.501	0.289	0.000	0.251	0.502	0.751	1.000
input_18	8946.000	0.501	0.288	0.000	0.251	0.501	0.750	1.000
input_19	8946.000	0.499	0.289	0.000	0.249	0.499	0.749	1.000
input_20	8946.000	0.500	0.288	0.000	0.250	0.500	0.748	1.000
input_21	8946.000	0.499	0.289	0.000	0.247	0.499	0.750	1.000
input_22	8946.000	0.508	0.287	0.000	0.262	0.512	0.756	1.000
input_23	8946.000	0.500	0.289	0.000	0.250	0.500	0.750	1.000
input_24	8946.000	0.501	0.288	0.000	0.252	0.501	0.749	1.000
input_25	8946.000	0.507	0.287	0.000	0.260	0.510	0.755	1.000
input_26	8946.000	0.504	0.287	0.000	0.255	0.505	0.752	1.000
input_27	8946.000	0.496	0.287	0.000	0.248	0.496	0.745	1.000
CoolingObjective_K2m2	8946.000	6.893	0.949	3.606	6.227	6.834	7.508	10.948
DelP_Pa	8946.000	60.188	33.370	3.962	41.453	52.777	69.003	797.843

split counts

Fold	Fit	Tune	Scale	Cal.	Separate ID	Test	Unused
Supplied random	4830	805	1208	1208	0	895	0
Design random	4294	715	1074	1073	0	1790	0
x0 low	3434	573	859	858	1432	1790	0
x0 middle	3434	573	859	858	1432	1790	0
x0 high	3434	573	859	858	1432	1790	0
x6 low	3434	573	859	858	1432	1790	0
x6 middle	3434	573	859	858	1432	1790	0
x6 high	3434	573	859	858	1432	1790	0
x12 low	3434	573	859	858	1432	1790	0
x12 middle	3434	573	859	858	1432	1790	0
x12 high	3434	573	859	858	1432	1790	0
Joint low	2751	458	688	688	1147	366	2848
Joint high	2749	458	688	687	1146	362	2856

Table 2: Exact row counts, identical across outer seeds. Random tests use their test set as ID evaluation; structured tests have a separate supported ID set. Joint stress tests deliberately exclude unused marginal-only cases.

full primary

Table 3: U-bend U+D-max results, means over three full runs. Empty accepted sets have undefined retained metrics. All error columns are percentages.

Fold	Target	Model	All MAPE %	Coverage %	Ret. MAPE %	Ret. fail %
Joint high	Cooling objective	CQR	5.118	51.934	4.752	40.449
Joint high	Cooling objective	ET-bag	6.416	52.118	5.992	48.281
Joint high	Cooling objective	ET-seed	6.282	53.039	5.871	48.059
Joint high	Cooling objective	GP-ARD	3.409	27.901	2.779	16.159
Joint high	Cooling objective	Log-poly	3.632	41.344	3.323	24.530
Joint high	Cooling objective	XGB	5.380	51.842	5.098	44.213
Joint high	Pressure drop	CQR	10.338	52.026	9.567	39.323
Joint high	Pressure drop	ET-bag	13.280	47.606	12.856	51.749
Joint high	Pressure drop	ET-seed	12.656	51.289	12.182	50.480
Joint high	Pressure drop	GP-ARD	10.175	32.505	9.537	39.242
Joint high	Pressure drop	Log-poly	18.111	41.344	15.916	61.782
Joint high	Pressure drop	XGB	10.258	53.407	9.300	36.020
Joint low	Cooling objective	CQR	5.177	52.914	5.039	41.972
Joint low	Cooling objective	ET-bag	5.742	51.548	5.417	44.198
Joint low	Cooling objective	ET-seed	5.845	54.007	5.580	47.183
Joint low	Cooling objective	GP-ARD	3.651	41.894	3.281	23.304
Joint low	Cooling objective	Log-poly	4.070	45.811	3.826	31.220
Joint low	Cooling objective	XGB	5.049	51.821	4.831	38.341
Joint low	Pressure drop	CQR	24.598	57.104	22.916	82.167
Joint low	Pressure drop	ET-bag	28.718	54.645	27.449	89.854
Joint low	Pressure drop	ET-seed	28.547	55.556	27.314	89.829
Joint low	Pressure drop	GP-ARD	13.098	38.160	11.403	45.301
Joint low	Pressure drop	Log-poly	15.570	45.811	14.078	57.516
Joint low	Pressure drop	XGB	23.292	53.825	22.450	84.087
Design random	Cooling objective	CQR	3.494	88.994	3.284	19.961
Design random	Cooling objective	ET-bag	4.974	89.423	4.805	37.636
Design random	Cooling objective	ET-seed	4.865	89.050	4.713	36.872
Design random	Cooling objective	GP-ARD	2.325	89.851	2.204	7.504
Design random	Cooling objective	Log-poly	2.554	90.354	2.480	10.634
Design random	Cooling objective	XGB	3.489	89.460	3.360	21.371
Design random	Pressure drop	CQR	10.825	90.149	10.108	39.901
Design random	Pressure drop	ET-bag	14.464	89.739	13.976	55.570
Design random	Pressure drop	ET-seed	13.769	89.609	13.327	53.233
Design random	Pressure drop	GP-ARD	7.583	89.777	7.292	25.863
Design random	Pressure drop	Log-poly	10.481	90.354	10.173	40.499
Design random	Pressure drop	XGB	10.333	89.181	9.750	38.818
x0 high	Cooling objective	CQR	3.350	77.281	3.134	19.447
x0 high	Cooling objective	ET-bag	4.416	77.747	4.215	31.440
x0 high	Cooling objective	ET-seed	4.324	79.218	4.149	30.523

Fold	Target	Model	All MAPE %	Coverage %	Ret. MAPE %	Ret. fail %
x0 high	Cooling objective	GP-ARD	2.779	66.555	2.462	10.021
x0 high	Cooling objective	Log-poly	3.027	71.657	2.773	14.449
x0 high	Cooling objective	XGB	3.261	79.590	3.120	19.572
x0 high	Pressure drop	CQR	9.481	77.635	8.810	33.364
x0 high	Pressure drop	ET-bag	12.006	77.747	11.440	44.314
x0 high	Pressure drop	ET-seed	11.384	78.994	10.814	41.480
x0 high	Pressure drop	GP-ARD	9.180	52.272	8.127	30.948
x0 high	Pressure drop	Log-poly	14.751	71.657	13.428	52.131
x0 high	Pressure drop	XGB	9.037	79.367	8.614	32.546
x0 low	Cooling objective	CQR	6.315	74.953	5.886	47.040
x0 low	Cooling objective	ET-bag	7.808	76.946	7.308	56.098
x0 low	Cooling objective	ET-seed	7.773	77.114	7.357	56.135
x0 low	Cooling objective	GP-ARD	3.399	60.819	3.000	17.794
x0 low	Cooling objective	Log-poly	3.548	71.750	3.365	22.263
x0 low	Cooling objective	XGB	6.245	75.363	5.896	47.722
x0 low	Pressure drop	CQR	19.201	73.780	17.682	66.655
x0 low	Pressure drop	ET-bag	21.926	77.169	21.120	74.485
x0 low	Pressure drop	ET-seed	21.475	75.158	20.719	73.168
x0 low	Pressure drop	GP-ARD	10.652	59.851	9.944	39.619
x0 low	Pressure drop	Log-poly	14.060	71.750	13.376	56.438
x0 low	Pressure drop	XGB	18.066	72.980	17.371	65.795
x0 middle	Cooling objective	CQR	3.622	92.756	3.487	23.709
x0 middle	Cooling objective	ET-bag	5.133	92.272	5.063	44.742
x0 middle	Cooling objective	ET-seed	5.022	93.222	4.960	43.083
x0 middle	Cooling objective	GP-ARD	2.282	94.302	2.239	8.293
x0 middle	Cooling objective	Log-poly	2.476	93.687	2.442	10.360
x0 middle	Cooling objective	XGB	3.656	93.818	3.573	24.629
x0 middle	Pressure drop	CQR	11.026	92.626	10.435	42.513
x0 middle	Pressure drop	ET-bag	16.283	92.607	15.896	62.955
x0 middle	Pressure drop	ET-seed	15.406	92.123	15.055	60.481
x0 middle	Pressure drop	GP-ARD	7.579	93.724	7.452	27.143
x0 middle	Pressure drop	Log-poly	10.102	93.687	9.905	38.780
x0 middle	Pressure drop	XGB	10.988	93.799	10.717	44.590
x12 high	Cooling objective	CQR	3.789	75.568	3.435	22.011
x12 high	Cooling objective	ET-bag	5.092	76.629	4.710	36.435
x12 high	Cooling objective	ET-seed	4.960	75.661	4.614	35.248
x12 high	Cooling objective	GP-ARD	2.524	75.531	2.308	9.016
x12 high	Cooling objective	Log-poly	2.753	70.782	2.535	10.392
x12 high	Cooling objective	XGB	3.711	74.842	3.395	21.720
x12 high	Pressure drop	CQR	11.441	74.935	9.362	36.598
x12 high	Pressure drop	ET-bag	14.284	75.419	12.515	49.543
x12 high	Pressure drop	ET-seed	13.684	75.475	12.080	47.083
x12 high	Pressure drop	GP-ARD	8.264	75.493	6.945	23.557
x12 high	Pressure drop	Log-poly	12.258	70.782	10.532	44.162
x12 high	Pressure drop	XGB	11.078	76.667	9.541	36.940
x12 low	Cooling objective	CQR	3.612	74.302	3.370	20.963

Fold	Target	Model	All MAPE %	Coverage %	Ret. MAPE %	Ret. fail %
x12 low	Cooling objective	ET-bag	5.055	74.134	4.787	38.171
x12 low	Cooling objective	ET-seed	4.917	75.829	4.719	37.129
x12 low	Cooling objective	GP-ARD	2.545	74.004	2.356	9.208
x12 low	Cooling objective	Log-poly	2.700	70.000	2.511	11.389
x12 low	Cooling objective	XGB	3.626	75.773	3.504	23.302
x12 low	Pressure drop	CQR	11.103	72.849	10.237	40.834
x12 low	Pressure drop	ET-bag	15.001	74.562	14.622	57.127
x12 low	Pressure drop	ET-seed	14.168	74.022	13.763	54.137
x12 low	Pressure drop	GP-ARD	7.768	74.600	7.366	26.379
x12 low	Pressure drop	Log-poly	10.470	70.000	9.818	38.915
x12 low	Pressure drop	XGB	10.811	73.892	10.354	42.829
x12 middle	Cooling objective	CQR	3.554	92.142	3.362	22.208
x12 middle	Cooling objective	ET-bag	4.968	91.899	4.832	38.953
x12 middle	Cooling objective	ET-seed	4.818	92.551	4.722	37.191
x12 middle	Cooling objective	GP-ARD	2.316	92.700	2.236	7.757
x12 middle	Cooling objective	Log-poly	2.492	93.966	2.442	10.641
x12 middle	Cooling objective	XGB	3.545	92.570	3.429	22.307
x12 middle	Pressure drop	CQR	10.937	91.806	10.242	40.547
x12 middle	Pressure drop	ET-bag	14.805	93.017	14.231	55.542
x12 middle	Pressure drop	ET-seed	14.073	92.812	13.590	53.454
x12 middle	Pressure drop	GP-ARD	7.664	92.086	7.431	26.732
x12 middle	Pressure drop	Log-poly	10.520	93.966	10.276	40.108
x12 middle	Pressure drop	XGB	10.594	92.551	10.193	40.446
x6 high	Cooling objective	CQR	4.534	75.568	4.034	30.128
x6 high	Cooling objective	ET-bag	6.057	76.201	5.557	43.656
x6 high	Cooling objective	ET-seed	5.799	76.909	5.320	41.150
x6 high	Cooling objective	GP-ARD	2.675	69.683	2.432	10.133
x6 high	Cooling objective	Log-poly	2.774	71.173	2.657	13.211
x6 high	Cooling objective	XGB	4.490	76.127	4.192	32.030
x6 high	Pressure drop	CQR	12.038	75.531	11.190	45.069
x6 high	Pressure drop	ET-bag	16.928	76.592	16.502	63.678
x6 high	Pressure drop	ET-seed	16.051	75.568	15.693	61.114
x6 high	Pressure drop	GP-ARD	7.723	74.655	7.224	25.644
x6 high	Pressure drop	Log-poly	11.263	71.173	10.506	41.655
x6 high	Pressure drop	XGB	11.712	75.922	11.146	45.148
x6 low	Cooling objective	CQR	4.727	76.313	4.517	39.339
x6 low	Cooling objective	ET-bag	6.514	74.823	6.550	60.641
x6 low	Cooling objective	ET-seed	6.209	77.132	6.205	57.173
x6 low	Cooling objective	GP-ARD	2.471	77.486	2.318	8.490
x6 low	Cooling objective	Log-poly	2.533	73.501	2.438	10.554
x6 low	Cooling objective	XGB	4.644	76.629	4.561	39.293
x6 low	Pressure drop	CQR	10.887	79.143	10.076	39.249
x6 low	Pressure drop	ET-bag	13.079	78.119	12.286	47.628
x6 low	Pressure drop	ET-seed	12.592	78.864	11.839	46.683
x6 low	Pressure drop	GP-ARD	8.006	77.114	7.376	26.474
x6 low	Pressure drop	Log-poly	10.487	73.501	9.830	41.361

Fold	Target	Model	All MAPE %	Coverage %	Ret. MAPE %	Ret. fail %
x6 low	Pressure drop	XGB	10.178	78.603	9.570	37.267
x6 middle	Cooling objective	CQR	3.704	94.898	3.574	23.469
x6 middle	Cooling objective	ET-bag	4.983	95.400	4.861	38.099
x6 middle	Cooling objective	ET-seed	4.857	89.870	4.704	36.876
x6 middle	Cooling objective	GP-ARD	2.356	94.953	2.291	8.815
x6 middle	Cooling objective	Log-poly	2.626	94.153	2.516	10.979
x6 middle	Cooling objective	XGB	3.806	91.862	3.704	25.203
x6 middle	Pressure drop	CQR	11.465	93.017	10.474	38.700
x6 middle	Pressure drop	ET-bag	15.074	93.296	14.378	54.186
x6 middle	Pressure drop	ET-seed	14.251	92.309	13.624	51.420
x6 middle	Pressure drop	GP-ARD	8.301	93.445	7.574	25.193
x6 middle	Pressure drop	Log-poly	11.375	94.153	10.161	40.248
x6 middle	Pressure drop	XGB	11.066	92.477	10.093	37.413
Supplied random	Cooling objective	CQR	3.336	89.758	3.170	18.579
Supplied random	Cooling objective	ET-bag	4.758	90.205	4.653	38.933
Supplied random	Cooling objective	ET-seed	4.624	89.944	4.544	37.342
Supplied random	Cooling objective	GP-ARD	2.285	90.801	2.174	7.923
Supplied random	Cooling objective	Log-poly	2.498	90.056	2.422	10.212
Supplied random	Cooling objective	XGB	3.275	90.056	3.178	19.679
Supplied random	Pressure drop	CQR	10.305	90.652	9.565	38.293
Supplied random	Pressure drop	ET-bag	13.905	90.056	13.454	54.259
Supplied random	Pressure drop	ET-seed	13.218	90.279	12.747	51.195
Supplied random	Pressure drop	GP-ARD	7.379	90.279	7.052	25.109
Supplied random	Pressure drop	Log-poly	9.826	90.056	9.557	38.374
Supplied random	Pressure drop	XGB	9.522	90.540	9.132	36.282

grouped

Target	Model	All MAPE %	Coverage %	Ret. MAPE %	Ret. fail %	All SD	Ret. SD
Cooling objective	CQR	3.494	88.994	3.284	19.961	0.057	0.060
Cooling objective	ET-bag	4.974	89.423	4.805	37.636	0.083	0.120
Cooling objective	ET-seed	4.865	89.050	4.713	36.872	0.081	0.133
Cooling objective	GP-ARD	2.325	89.851	2.204	7.504	0.008	0.015
Cooling objective	Log-poly	2.554	90.354	2.480	10.634	0.018	0.046
Cooling objective	XGB	3.489	89.460	3.360	21.371	0.049	0.043
Pressure drop	CQR	10.825	90.149	10.108	39.901	0.227	0.221
Pressure drop	ET-bag	14.464	89.739	13.976	55.570	0.228	0.056
Pressure drop	ET-seed	13.769	89.609	13.327	53.233	0.247	0.319
Pressure drop	GP-ARD	7.583	89.777	7.292	25.863	0.182	0.079
Pressure drop	Log-poly	10.481	90.354	10.173	40.499	0.210	0.183
Pressure drop	XGB	10.333	89.181	9.750	38.818	0.112	0.074

Table 4: Random-design evaluation, three-run means and sample SD. Calibration uses a nominal 90% acceptance budget; measured test coverage is shown.

supplied random

Target	Model	All MAPE %	Coverage %	Ret. MAPE %	Ret. fail %	All SD	Ret. SD
Cooling objective	CQR	3.336	89.758	3.170	18.579	0.047	0.069
Cooling objective	ET-bag	4.758	90.205	4.653	38.933	0.052	0.089
Cooling objective	ET-seed	4.624	89.944	4.544	37.342	0.045	0.061
Cooling objective	GP-ARD	2.285	90.801	2.174	7.923	0.030	0.046
Cooling objective	Log-poly	2.498	90.056	2.422	10.212	0.028	0.040
Cooling objective	XGB	3.275	90.056	3.178	19.679	0.060	0.072
Pressure drop	CQR	10.305	90.652	9.565	38.293	0.179	0.071
Pressure drop	ET-bag	13.905	90.056	13.454	54.259	0.130	0.059
Pressure drop	ET-seed	13.218	90.279	12.747	51.195	0.154	0.092
Pressure drop	GP-ARD	7.379	90.279	7.052	25.109	0.586	0.588
Pressure drop	Log-poly	9.826	90.056	9.557	38.374	0.168	0.219
Pressure drop	XGB	9.522	90.540	9.132	36.282	0.191	0.164

Table 5: Random-design evaluation, three-run means and sample SD. Calibration uses a nominal 90% acceptance budget; measured test coverage is shown.

structured

Table 6: Geometry-coordinate support shifts, defined before examining errors; low/high are endpoint support tests and middle is a missing interior slab. Means over three runs.

Fold	Target	Model	All MAPE %	Coverage %	Ret. MAPE %	Ret. fail %
Joint high	Cooling objective	CQR	5.118	51.934	4.752	40.449
Joint high	Cooling objective	ET-bag	6.416	52.118	5.992	48.281
Joint high	Cooling objective	ET-seed	6.282	53.039	5.871	48.059
Joint high	Cooling objective	GP-ARD	3.409	27.901	2.779	16.159
Joint high	Cooling objective	Log-poly	3.632	41.344	3.323	24.530
Joint high	Cooling objective	XGB	5.380	51.842	5.098	44.213
Joint high	Pressure drop	CQR	10.338	52.026	9.567	39.323
Joint high	Pressure drop	ET-bag	13.280	47.606	12.856	51.749
Joint high	Pressure drop	ET-seed	12.656	51.289	12.182	50.480
Joint high	Pressure drop	GP-ARD	10.175	32.505	9.537	39.242
Joint high	Pressure drop	Log-poly	18.111	41.344	15.916	61.782
Joint high	Pressure drop	XGB	10.258	53.407	9.300	36.020
Joint low	Cooling objective	CQR	5.177	52.914	5.039	41.972
Joint low	Cooling objective	ET-bag	5.742	51.548	5.417	44.198
Joint low	Cooling objective	ET-seed	5.845	54.007	5.580	47.183
Joint low	Cooling objective	GP-ARD	3.651	41.894	3.281	23.304
Joint low	Cooling objective	Log-poly	4.070	45.811	3.826	31.220
Joint low	Cooling objective	XGB	5.049	51.821	4.831	38.341
Joint low	Pressure drop	CQR	24.598	57.104	22.916	82.167
Joint low	Pressure drop	ET-bag	28.718	54.645	27.449	89.854
Joint low	Pressure drop	ET-seed	28.547	55.556	27.314	89.829
Joint low	Pressure drop	GP-ARD	13.098	38.160	11.403	45.301
Joint low	Pressure drop	Log-poly	15.570	45.811	14.078	57.516
Joint low	Pressure drop	XGB	23.292	53.825	22.450	84.087
x0 high	Cooling objective	CQR	3.350	77.281	3.134	19.447
x0 high	Cooling objective	ET-bag	4.416	77.747	4.215	31.440
x0 high	Cooling objective	ET-seed	4.324	79.218	4.149	30.523
x0 high	Cooling objective	GP-ARD	2.779	66.555	2.462	10.021
x0 high	Cooling objective	Log-poly	3.027	71.657	2.773	14.449
x0 high	Cooling objective	XGB	3.261	79.590	3.120	19.572
x0 high	Pressure drop	CQR	9.481	77.635	8.810	33.364
x0 high	Pressure drop	ET-bag	12.006	77.747	11.440	44.314
x0 high	Pressure drop	ET-seed	11.384	78.994	10.814	41.480
x0 high	Pressure drop	GP-ARD	9.180	52.272	8.127	30.948
x0 high	Pressure drop	Log-poly	14.751	71.657	13.428	52.131
x0 high	Pressure drop	XGB	9.037	79.367	8.614	32.546
x0 low	Cooling objective	CQR	6.315	74.953	5.886	47.040
x0 low	Cooling objective	ET-bag	7.808	76.946	7.308	56.098
x0 low	Cooling objective	ET-seed	7.773	77.114	7.357	56.135
x0 low	Cooling objective	GP-ARD	3.399	60.819	3.000	17.794

Fold	Target	Model	All MAPE %	Coverage %	Ret. MAPE %	Ret. fail %
x0 low	Cooling objective	Log-poly	3.548	71.750	3.365	22.263
x0 low	Cooling objective	XGB	6.245	75.363	5.896	47.722
x0 low	Pressure drop	CQR	19.201	73.780	17.682	66.655
x0 low	Pressure drop	ET-bag	21.926	77.169	21.120	74.485
x0 low	Pressure drop	ET-seed	21.475	75.158	20.719	73.168
x0 low	Pressure drop	GP-ARD	10.652	59.851	9.944	39.619
x0 low	Pressure drop	Log-poly	14.060	71.750	13.376	56.438
x0 low	Pressure drop	XGB	18.066	72.980	17.371	65.795
x0 middle	Cooling objective	CQR	3.622	92.756	3.487	23.709
x0 middle	Cooling objective	ET-bag	5.133	92.272	5.063	44.742
x0 middle	Cooling objective	ET-seed	5.022	93.222	4.960	43.083
x0 middle	Cooling objective	GP-ARD	2.282	94.302	2.239	8.293
x0 middle	Cooling objective	Log-poly	2.476	93.687	2.442	10.360
x0 middle	Cooling objective	XGB	3.656	93.818	3.573	24.629
x0 middle	Pressure drop	CQR	11.026	92.626	10.435	42.513
x0 middle	Pressure drop	ET-bag	16.283	92.607	15.896	62.955
x0 middle	Pressure drop	ET-seed	15.406	92.123	15.055	60.481
x0 middle	Pressure drop	GP-ARD	7.579	93.724	7.452	27.143
x0 middle	Pressure drop	Log-poly	10.102	93.687	9.905	38.780
x0 middle	Pressure drop	XGB	10.988	93.799	10.717	44.590
x12 high	Cooling objective	CQR	3.789	75.568	3.435	22.011
x12 high	Cooling objective	ET-bag	5.092	76.629	4.710	36.435
x12 high	Cooling objective	ET-seed	4.960	75.661	4.614	35.248
x12 high	Cooling objective	GP-ARD	2.524	75.531	2.308	9.016
x12 high	Cooling objective	Log-poly	2.753	70.782	2.535	10.392
x12 high	Cooling objective	XGB	3.711	74.842	3.395	21.720
x12 high	Pressure drop	CQR	11.441	74.935	9.362	36.598
x12 high	Pressure drop	ET-bag	14.284	75.419	12.515	49.543
x12 high	Pressure drop	ET-seed	13.684	75.475	12.080	47.083
x12 high	Pressure drop	GP-ARD	8.264	75.493	6.945	23.557
x12 high	Pressure drop	Log-poly	12.258	70.782	10.532	44.162
x12 high	Pressure drop	XGB	11.078	76.667	9.541	36.940
x12 low	Cooling objective	CQR	3.612	74.302	3.370	20.963
x12 low	Cooling objective	ET-bag	5.055	74.134	4.787	38.171
x12 low	Cooling objective	ET-seed	4.917	75.829	4.719	37.129
x12 low	Cooling objective	GP-ARD	2.545	74.004	2.356	9.208
x12 low	Cooling objective	Log-poly	2.700	70.000	2.511	11.389
x12 low	Cooling objective	XGB	3.626	75.773	3.504	23.302
x12 low	Pressure drop	CQR	11.103	72.849	10.237	40.834
x12 low	Pressure drop	ET-bag	15.001	74.562	14.622	57.127
x12 low	Pressure drop	ET-seed	14.168	74.022	13.763	54.137
x12 low	Pressure drop	GP-ARD	7.768	74.600	7.366	26.379
x12 low	Pressure drop	Log-poly	10.470	70.000	9.818	38.915
x12 low	Pressure drop	XGB	10.811	73.892	10.354	42.829
x12 middle	Cooling objective	CQR	3.554	92.142	3.362	22.208
x12 middle	Cooling objective	ET-bag	4.968	91.899	4.832	38.953

Fold	Target	Model	All MAPE %	Coverage %	Ret. MAPE %	Ret. fail %
x12 middle	Cooling objective	ET-seed	4.818	92.551	4.722	37.191
x12 middle	Cooling objective	GP-ARD	2.316	92.700	2.236	7.757
x12 middle	Cooling objective	Log-poly	2.492	93.966	2.442	10.641
x12 middle	Cooling objective	XGB	3.545	92.570	3.429	22.307
x12 middle	Pressure drop	CQR	10.937	91.806	10.242	40.547
x12 middle	Pressure drop	ET-bag	14.805	93.017	14.231	55.542
x12 middle	Pressure drop	ET-seed	14.073	92.812	13.590	53.454
x12 middle	Pressure drop	GP-ARD	7.664	92.086	7.431	26.732
x12 middle	Pressure drop	Log-poly	10.520	93.966	10.276	40.108
x12 middle	Pressure drop	XGB	10.594	92.551	10.193	40.446
x6 high	Cooling objective	CQR	4.534	75.568	4.034	30.128
x6 high	Cooling objective	ET-bag	6.057	76.201	5.557	43.656
x6 high	Cooling objective	ET-seed	5.799	76.909	5.320	41.150
x6 high	Cooling objective	GP-ARD	2.675	69.683	2.432	10.133
x6 high	Cooling objective	Log-poly	2.774	71.173	2.657	13.211
x6 high	Cooling objective	XGB	4.490	76.127	4.192	32.030
x6 high	Pressure drop	CQR	12.038	75.531	11.190	45.069
x6 high	Pressure drop	ET-bag	16.928	76.592	16.502	63.678
x6 high	Pressure drop	ET-seed	16.051	75.568	15.693	61.114
x6 high	Pressure drop	GP-ARD	7.723	74.655	7.224	25.644
x6 high	Pressure drop	Log-poly	11.263	71.173	10.506	41.655
x6 high	Pressure drop	XGB	11.712	75.922	11.146	45.148
x6 low	Cooling objective	CQR	4.727	76.313	4.517	39.339
x6 low	Cooling objective	ET-bag	6.514	74.823	6.550	60.641
x6 low	Cooling objective	ET-seed	6.209	77.132	6.205	57.173
x6 low	Cooling objective	GP-ARD	2.471	77.486	2.318	8.490
x6 low	Cooling objective	Log-poly	2.533	73.501	2.438	10.554
x6 low	Cooling objective	XGB	4.644	76.629	4.561	39.293
x6 low	Pressure drop	CQR	10.887	79.143	10.076	39.249
x6 low	Pressure drop	ET-bag	13.079	78.119	12.286	47.628
x6 low	Pressure drop	ET-seed	12.592	78.864	11.839	46.683
x6 low	Pressure drop	GP-ARD	8.006	77.114	7.376	26.474
x6 low	Pressure drop	Log-poly	10.487	73.501	9.830	41.361
x6 low	Pressure drop	XGB	10.178	78.603	9.570	37.267
x6 middle	Cooling objective	CQR	3.704	94.898	3.574	23.469
x6 middle	Cooling objective	ET-bag	4.983	95.400	4.861	38.099
x6 middle	Cooling objective	ET-seed	4.857	89.870	4.704	36.876
x6 middle	Cooling objective	GP-ARD	2.356	94.953	2.291	8.815
x6 middle	Cooling objective	Log-poly	2.626	94.153	2.516	10.979
x6 middle	Cooling objective	XGB	3.806	91.862	3.704	25.203
x6 middle	Pressure drop	CQR	11.465	93.017	10.474	38.700
x6 middle	Pressure drop	ET-bag	15.074	93.296	14.378	54.186
x6 middle	Pressure drop	ET-seed	14.251	92.309	13.624	51.420
x6 middle	Pressure drop	GP-ARD	8.301	93.445	7.574	25.193
x6 middle	Pressure drop	Log-poly	11.375	94.153	10.161	40.248
x6 middle	Pressure drop	XGB	11.066	92.477	10.093	37.413

gate decisions

Table 7: Binary calibrated acceptance/abstention. The warning tier is a separate retrospective diagnostic only.

Fold	Target	Model	All MAPE %	Coverage %	Ret. MAPE %	Ret. fail %	Abstained %
Joint high	Cooling objective	ET-seed	6.282	53.039	5.871	48.059	46.961
Joint high	Cooling objective	GP-ARD	3.409	27.901	2.779	16.159	72.099
Joint high	Cooling objective	XGB	5.380	51.842	5.098	44.213	48.158
Joint high	Pressure drop	ET-seed	12.656	51.289	12.182	50.480	48.711
Joint high	Pressure drop	GP-ARD	10.175	32.505	9.537	39.242	67.495
Joint high	Pressure drop	XGB	10.258	53.407	9.300	36.020	46.593
Design random	Cooling objective	ET-seed	4.865	89.050	4.713	36.872	10.950
Design random	Cooling objective	GP-ARD	2.325	89.851	2.204	7.504	10.149
Design random	Cooling objective	XGB	3.489	89.460	3.360	21.371	10.540
Design random	Pressure drop	ET-seed	13.769	89.609	13.327	53.233	10.391
Design random	Pressure drop	GP-ARD	7.583	89.777	7.292	25.863	10.223
Design random	Pressure drop	XGB	10.333	89.181	9.750	38.818	10.819
x0 high	Cooling objective	ET-seed	4.324	79.218	4.149	30.523	20.782
x0 high	Cooling objective	GP-ARD	2.779	66.555	2.462	10.021	33.445
x0 high	Cooling objective	XGB	3.261	79.590	3.120	19.572	20.410
x0 high	Pressure drop	ET-seed	11.384	78.994	10.814	41.480	21.006
x0 high	Pressure drop	GP-ARD	9.180	52.272	8.127	30.948	47.728
x0 high	Pressure drop	XGB	9.037	79.367	8.614	32.546	20.633
x0 low	Cooling objective	ET-seed	7.773	77.114	7.357	56.135	22.886
x0 low	Cooling objective	GP-ARD	3.399	60.819	3.000	17.794	39.181
x0 low	Cooling objective	XGB	6.245	75.363	5.896	47.722	24.637
x0 low	Pressure drop	ET-seed	21.475	75.158	20.719	73.168	24.842
x0 low	Pressure drop	GP-ARD	10.652	59.851	9.944	39.619	40.149
x0 low	Pressure drop	XGB	18.066	72.980	17.371	65.795	27.020
x0 middle	Cooling objective	ET-seed	5.022	93.222	4.960	43.083	6.778
x0 middle	Cooling objective	GP-ARD	2.282	94.302	2.239	8.293	5.698
x0 middle	Cooling objective	XGB	3.656	93.818	3.573	24.629	6.182
x0 middle	Pressure drop	ET-seed	15.406	92.123	15.055	60.481	7.877
x0 middle	Pressure drop	GP-ARD	7.579	93.724	7.452	27.143	6.276
x0 middle	Pressure drop	XGB	10.988	93.799	10.717	44.590	6.201
x6 high	Cooling objective	ET-seed	5.799	76.909	5.320	41.150	23.091
x6 high	Cooling objective	GP-ARD	2.675	69.683	2.432	10.133	30.317
x6 high	Cooling objective	XGB	4.490	76.127	4.192	32.030	23.873
x6 high	Pressure drop	ET-seed	16.051	75.568	15.693	61.114	24.432
x6 high	Pressure drop	GP-ARD	7.723	74.655	7.224	25.644	25.345
x6 high	Pressure drop	XGB	11.712	75.922	11.146	45.148	24.078

false decisions

Table 8: False accept and false abstain use all test rows as denominator; failure prevalence is tolerance defined. Tolerances are illustrative, not engineering standards.

Fold	Target	Model	Failures %	False accept %	False abstain %
Joint high	Cooling objective	ET-seed	52.026	25.414	20.350
Joint high	Cooling objective	GP-ARD	24.217	4.420	52.302
Joint high	Cooling objective	XGB	47.514	22.928	23.573
Joint high	Pressure drop	ET-seed	50.460	25.875	24.125
Joint high	Pressure drop	GP-ARD	41.713	12.891	38.674
Joint high	Pressure drop	XGB	40.147	19.153	25.599
Design random	Cooling objective	ET-seed	37.561	32.831	6.220
Design random	Cooling objective	GP-ARD	9.125	6.741	7.765
Design random	Cooling objective	XGB	22.607	19.125	7.058
Design random	Pressure drop	ET-seed	54.078	47.709	4.022
Design random	Pressure drop	GP-ARD	27.486	23.222	5.959
Design random	Pressure drop	XGB	40.615	34.618	4.823
x0 high	Cooling objective	ET-seed	32.216	24.171	12.737
x0 high	Cooling objective	GP-ARD	14.134	6.648	25.959
x0 high	Cooling objective	XGB	20.968	15.568	15.009
x0 high	Pressure drop	ET-seed	43.333	32.756	10.428
x0 high	Pressure drop	GP-ARD	35.978	16.164	27.914
x0 high	Pressure drop	XGB	34.283	25.829	12.179
x0 low	Cooling objective	ET-seed	58.119	43.296	8.063
x0 low	Cooling objective	GP-ARD	22.868	10.838	27.151
x0 low	Cooling objective	XGB	50.186	35.959	10.410
x0 low	Pressure drop	ET-seed	74.562	55.009	5.289
x0 low	Pressure drop	GP-ARD	42.775	23.631	21.006
x0 low	Pressure drop	XGB	67.318	48.063	7.765
x0 middle	Cooling objective	ET-seed	43.333	40.168	3.613
x0 middle	Cooling objective	GP-ARD	8.957	7.821	4.562
x0 middle	Cooling objective	XGB	25.456	23.110	3.836
x0 middle	Pressure drop	ET-seed	60.968	55.717	2.626
x0 middle	Pressure drop	GP-ARD	27.933	25.438	3.780
x0 middle	Pressure drop	XGB	45.214	41.844	2.831
x6 high	Cooling objective	ET-seed	43.892	31.676	10.875
x6 high	Cooling objective	GP-ARD	13.464	7.058	23.911
x6 high	Cooling objective	XGB	34.134	24.376	14.115
x6 high	Pressure drop	ET-seed	60.857	46.164	9.739
x6 high	Pressure drop	GP-ARD	28.082	19.143	16.406
x6 high	Pressure drop	XGB	46.723	34.283	11.639

detection

Table 9: Failure detection means over evaluable runs; undefined single-class detection metrics remain blank/–. Valid AUC run count is explicit.

Fold	Target	Model	Failures %	AUROC	AUPRC	Spearman	Valid AUC runs
Joint high	Cooling objective	CQR	43.554	0.568	0.509	0.160	3
Joint high	Cooling objective	ET-bag	52.855	0.585	0.614	0.125	3
Joint high	Cooling objective	ET-seed	52.026	0.577	0.614	0.125	3
Joint high	Cooling objective	GP-ARD	24.217	0.615	0.348	0.178	3
Joint high	Cooling objective	Log-poly	26.059	0.552	0.313	0.114	3
Joint high	Cooling objective	XGB	47.514	0.559	0.535	0.119	3
Joint high	Pressure drop	CQR	42.726	0.544	0.464	0.090	3
Joint high	Pressure drop	ET-bag	53.499	0.521	0.545	0.044	3
Joint high	Pressure drop	ET-seed	50.460	0.517	0.511	0.052	3
Joint high	Pressure drop	GP-ARD	41.713	0.546	0.480	0.065	3
Joint high	Pressure drop	Log-poly	63.536	0.536	0.685	0.109	3
Joint high	Pressure drop	XGB	40.147	0.570	0.452	0.122	3
Joint low	Cooling objective	CQR	42.623	0.511	0.449	0.037	3
Joint low	Cooling objective	ET-bag	45.082	0.500	0.454	0.024	3
Joint low	Cooling objective	ET-seed	47.177	0.483	0.458	0.014	3
Joint low	Cooling objective	GP-ARD	28.051	0.573	0.343	0.112	3
Joint low	Cooling objective	Log-poly	32.605	0.520	0.348	0.050	3
Joint low	Cooling objective	XGB	39.435	0.523	0.419	0.054	3
Joint low	Pressure drop	CQR	84.699	0.563	0.880	0.130	3
Joint low	Pressure drop	ET-bag	90.893	0.546	0.923	0.081	3
Joint low	Pressure drop	ET-seed	91.166	0.554	0.927	0.084	3
Joint low	Pressure drop	GP-ARD	54.189	0.587	0.616	0.157	3
Joint low	Pressure drop	Log-poly	62.477	0.553	0.679	0.122	3
Joint low	Pressure drop	XGB	83.060	0.482	0.835	0.058	3
Design random	Cooling objective	CQR	22.086	0.633	0.347	0.172	3
Design random	Cooling objective	ET-bag	38.622	0.558	0.438	0.087	3
Design random	Cooling objective	ET-seed	37.561	0.542	0.407	0.086	3
Design random	Cooling objective	GP-ARD	9.125	0.680	0.197	0.184	3
Design random	Cooling objective	Log-poly	11.769	0.606	0.180	0.086	3
Design random	Cooling objective	XGB	22.607	0.584	0.297	0.107	3
Design random	Pressure drop	CQR	41.285	0.575	0.496	0.163	3
Design random	Pressure drop	ET-bag	56.425	0.529	0.597	0.088	3
Design random	Pressure drop	ET-seed	54.078	0.541	0.583	0.105	3
Design random	Pressure drop	GP-ARD	27.486	0.598	0.359	0.151	3
Design random	Pressure drop	Log-poly	41.341	0.547	0.457	0.113	3
Design random	Pressure drop	XGB	40.615	0.583	0.489	0.153	3
x0 high	Cooling objective	CQR	21.732	0.598	0.293	0.148	3
x0 high	Cooling objective	ET-bag	32.980	0.558	0.379	0.112	3
x0 high	Cooling objective	ET-seed	32.216	0.561	0.377	0.108	3
x0 high	Cooling objective	GP-ARD	14.134	0.664	0.244	0.195	3

Fold	Target	Model	Failures %	AUROC	AUPRC	Spearman	Valid AUC runs
x0 high	Cooling objective	Log-poly	17.784	0.610	0.256	0.138	3
x0 high	Cooling objective	XGB	20.968	0.575	0.270	0.125	3
x0 high	Pressure drop	CQR	35.885	0.574	0.431	0.132	3
x0 high	Pressure drop	ET-bag	45.475	0.542	0.493	0.092	3
x0 high	Pressure drop	ET-seed	43.333	0.557	0.488	0.107	3
x0 high	Pressure drop	GP-ARD	35.978	0.591	0.449	0.163	3
x0 high	Pressure drop	Log-poly	54.618	0.554	0.599	0.140	3
x0 high	Pressure drop	XGB	34.283	0.562	0.399	0.110	3
x0 low	Cooling objective	CQR	49.963	0.577	0.577	0.171	3
x0 low	Cooling objective	ET-bag	58.343	0.570	0.646	0.161	3
x0 low	Cooling objective	ET-seed	58.119	0.566	0.637	0.152	3
x0 low	Cooling objective	GP-ARD	22.868	0.617	0.308	0.159	3
x0 low	Cooling objective	Log-poly	24.264	0.568	0.296	0.095	3
x0 low	Cooling objective	XGB	50.186	0.567	0.564	0.138	3
x0 low	Pressure drop	CQR	69.050	0.567	0.743	0.190	3
x0 low	Pressure drop	ET-bag	75.810	0.555	0.793	0.133	3
x0 low	Pressure drop	ET-seed	74.562	0.539	0.769	0.119	3
x0 low	Pressure drop	GP-ARD	42.775	0.561	0.485	0.113	3
x0 low	Pressure drop	Log-poly	57.505	0.523	0.599	0.079	3
x0 low	Pressure drop	XGB	67.318	0.542	0.700	0.114	3
x0 middle	Cooling objective	CQR	25.307	0.610	0.355	0.151	3
x0 middle	Cooling objective	ET-bag	45.214	0.528	0.480	0.069	3
x0 middle	Cooling objective	ET-seed	43.333	0.524	0.454	0.066	3
x0 middle	Cooling objective	GP-ARD	8.957	0.660	0.158	0.154	3
x0 middle	Cooling objective	Log-poly	10.894	0.619	0.158	0.108	3
x0 middle	Cooling objective	XGB	25.456	0.569	0.309	0.096	3
x0 middle	Pressure drop	CQR	43.818	0.569	0.512	0.143	3
x0 middle	Pressure drop	ET-bag	63.352	0.484	0.640	0.023	3
x0 middle	Pressure drop	ET-seed	60.968	0.483	0.613	0.013	3
x0 middle	Pressure drop	GP-ARD	27.933	0.583	0.341	0.118	3
x0 middle	Pressure drop	Log-poly	39.572	0.550	0.450	0.094	3
x0 middle	Pressure drop	XGB	45.214	0.540	0.492	0.092	3
x12 high	Cooling objective	CQR	24.935	0.608	0.324	0.142	3
x12 high	Cooling objective	ET-bag	38.194	0.558	0.422	0.115	3
x12 high	Cooling objective	ET-seed	36.778	0.557	0.406	0.105	3
x12 high	Cooling objective	GP-ARD	11.099	0.634	0.170	0.128	3
x12 high	Cooling objective	Log-poly	12.831	0.615	0.198	0.074	3
x12 high	Cooling objective	XGB	24.358	0.597	0.306	0.124	3
x12 high	Pressure drop	CQR	40.168	0.585	0.493	0.166	3
x12 high	Pressure drop	ET-bag	51.993	0.548	0.568	0.127	3
x12 high	Pressure drop	ET-seed	49.441	0.550	0.548	0.119	3
x12 high	Pressure drop	GP-ARD	26.667	0.597	0.340	0.131	3
x12 high	Pressure drop	Log-poly	45.568	0.538	0.492	0.091	3
x12 high	Pressure drop	XGB	39.460	0.572	0.460	0.139	3
x12 low	Cooling objective	CQR	23.538	0.589	0.310	0.131	3
x12 low	Cooling objective	ET-bag	40.577	0.558	0.464	0.107	3

Fold	Target	Model	Failures %	AUROC	AUPRC	Spearman	Valid AUC runs
x12 low	Cooling objective	ET-seed	38.957	0.557	0.446	0.096	3
x12 low	Cooling objective	GP-ARD	11.844	0.641	0.196	0.161	3
x12 low	Cooling objective	Log-poly	13.873	0.620	0.216	0.133	3
x12 low	Cooling objective	XGB	24.637	0.570	0.299	0.091	3
x12 low	Pressure drop	CQR	43.333	0.564	0.509	0.126	3
x12 low	Pressure drop	ET-bag	57.225	0.504	0.581	0.037	3
x12 low	Pressure drop	ET-seed	54.898	0.520	0.571	0.059	3
x12 low	Pressure drop	GP-ARD	28.659	0.579	0.356	0.129	3
x12 low	Pressure drop	Log-poly	40.838	0.542	0.450	0.098	3
x12 low	Pressure drop	XGB	43.966	0.545	0.492	0.088	3
x12 middle	Cooling objective	CQR	23.911	0.607	0.348	0.164	3
x12 middle	Cooling objective	ET-bag	40.000	0.560	0.457	0.116	3
x12 middle	Cooling objective	ET-seed	37.728	0.549	0.418	0.094	3
x12 middle	Cooling objective	GP-ARD	8.901	0.695	0.188	0.185	3
x12 middle	Cooling objective	Log-poly	11.210	0.599	0.162	0.092	3
x12 middle	Cooling objective	XGB	23.184	0.573	0.291	0.112	3
x12 middle	Pressure drop	CQR	41.825	0.569	0.498	0.150	3
x12 middle	Pressure drop	ET-bag	56.331	0.539	0.609	0.104	3
x12 middle	Pressure drop	ET-seed	54.264	0.547	0.597	0.102	3
x12 middle	Pressure drop	GP-ARD	27.765	0.593	0.344	0.162	3
x12 middle	Pressure drop	Log-poly	40.726	0.551	0.455	0.125	3
x12 middle	Pressure drop	XGB	41.657	0.574	0.494	0.158	3
x6 high	Cooling objective	CQR	33.724	0.596	0.440	0.179	3
x6 high	Cooling objective	ET-bag	46.406	0.573	0.528	0.148	3
x6 high	Cooling objective	ET-seed	43.892	0.572	0.506	0.149	3
x6 high	Cooling objective	GP-ARD	13.464	0.644	0.229	0.175	3
x6 high	Cooling objective	Log-poly	14.320	0.566	0.191	0.056	3
x6 high	Cooling objective	XGB	34.134	0.560	0.391	0.129	3
x6 high	Pressure drop	CQR	46.965	0.550	0.521	0.110	3
x6 high	Pressure drop	ET-bag	63.762	0.494	0.637	0.023	3
x6 high	Pressure drop	ET-seed	60.857	0.495	0.610	0.025	3
x6 high	Pressure drop	GP-ARD	28.082	0.593	0.363	0.165	3
x6 high	Pressure drop	Log-poly	44.451	0.562	0.509	0.117	3
x6 high	Pressure drop	XGB	46.723	0.547	0.508	0.105	3
x6 low	Cooling objective	CQR	41.341	0.561	0.464	0.120	3
x6 low	Cooling objective	ET-bag	58.827	0.449	0.552	-0.052	3
x6 low	Cooling objective	ET-seed	56.071	0.470	0.539	-0.027	3
x6 low	Cooling objective	GP-ARD	10.652	0.651	0.185	0.161	3
x6 low	Cooling objective	Log-poly	11.844	0.573	0.154	0.090	3
x6 low	Cooling objective	XGB	40.056	0.517	0.419	0.030	3
x6 low	Pressure drop	CQR	42.142	0.578	0.505	0.143	3
x6 low	Pressure drop	ET-bag	49.739	0.557	0.553	0.125	3
x6 low	Pressure drop	ET-seed	48.547	0.554	0.538	0.119	3
x6 low	Pressure drop	GP-ARD	29.683	0.595	0.383	0.179	3
x6 low	Pressure drop	Log-poly	42.533	0.553	0.469	0.122	3
x6 low	Pressure drop	XGB	39.683	0.571	0.458	0.120	3

Fold	Target	Model	Failures %	AUROC	AUPRC	Spearman	Valid AUC runs
x6 middle	Cooling objective	CQR	24.507	0.587	0.330	0.128	3
x6 middle	Cooling objective	ET-bag	38.603	0.555	0.437	0.093	3
x6 middle	Cooling objective	ET-seed	37.467	0.546	0.410	0.074	3
x6 middle	Cooling objective	GP-ARD	9.404	0.672	0.183	0.160	3
x6 middle	Cooling objective	Log-poly	11.546	0.615	0.163	0.099	3
x6 middle	Cooling objective	XGB	25.922	0.560	0.304	0.088	3
x6 middle	Pressure drop	CQR	40.298	0.570	0.495	0.140	3
x6 middle	Pressure drop	ET-bag	55.233	0.545	0.609	0.132	3
x6 middle	Pressure drop	ET-seed	52.402	0.538	0.570	0.114	3
x6 middle	Pressure drop	GP-ARD	26.108	0.614	0.346	0.158	3
x6 middle	Pressure drop	Log-poly	41.061	0.567	0.473	0.129	3
x6 middle	Pressure drop	XGB	38.957	0.586	0.479	0.165	3
Supplied random	Cooling objective	CQR	20.559	0.596	0.314	0.107	3
Supplied random	Cooling objective	ET-bag	39.069	0.501	0.398	0.016	3
Supplied random	Cooling objective	ET-seed	37.281	0.489	0.368	0.002	3
Supplied random	Cooling objective	GP-ARD	9.199	0.654	0.193	0.177	3
Supplied random	Cooling objective	Log-poly	11.285	0.620	0.177	0.106	3
Supplied random	Cooling objective	XGB	20.447	0.559	0.246	0.045	3
Supplied random	Pressure drop	CQR	40.559	0.581	0.511	0.168	3
Supplied random	Pressure drop	ET-bag	54.898	0.519	0.583	0.084	3
Supplied random	Pressure drop	ET-seed	52.142	0.531	0.566	0.080	3
Supplied random	Pressure drop	GP-ARD	26.704	0.606	0.351	0.181	3
Supplied random	Pressure drop	Log-poly	39.143	0.557	0.438	0.112	3
Supplied random	Pressure drop	XGB	37.840	0.584	0.465	0.155	3

ablation

Score	Target	Coverage %	Ret. MAPE %	AURC	AUROC	AUPRC	Spearman
Distance	Cooling objective	90.354	3.396	3.109	0.575	0.279	0.098
Distance	Pressure drop	90.354	10.139	9.410	0.548	0.445	0.084
Uncertainty	Cooling objective	89.330	3.352	3.265	0.546	0.281	0.066
Uncertainty	Pressure drop	89.404	9.660	8.797	0.588	0.504	0.171
Equal U+D+V	Cooling objective	89.907	3.333	3.091	0.582	0.300	0.108
Equal U+D+V	Pressure drop	89.758	9.712	8.729	0.593	0.501	0.175
Max U+D+V	Cooling objective	89.460	3.360	3.106	0.584	0.297	0.107
Max U+D+V	Pressure drop	89.181	9.750	8.865	0.583	0.489	0.153
Equal U+D	Cooling objective	89.907	3.333	3.091	0.582	0.300	0.108
Equal U+D	Pressure drop	89.758	9.712	8.729	0.593	0.501	0.175
Max U+D	Cooling objective	89.460	3.360	3.106	0.584	0.297	0.107
Max U+D	Pressure drop	89.181	9.750	8.866	0.583	0.489	0.153
Range/positivity	Cooling objective	100.000	3.489	–	0.500	0.226	–
Range/positivity	Pressure drop	100.000	10.333	–	0.500	0.406	–
all	Cooling objective	100.000	3.489	–	0.500	0.226	–
all	Pressure drop	100.000	10.333	–	0.500	0.406	–
Learned gate	Cooling objective	90.279	3.341	3.077	0.587	0.298	0.111
Learned gate	Pressure drop	89.795	9.659	8.678	0.595	0.507	0.181
Clipped reference	Cooling objective	96.667	3.438	3.108	0.582	0.283	0.106
Clipped reference	Pressure drop	96.518	10.159	8.879	0.582	0.475	0.150
Leverage	Cooling objective	90.074	3.395	3.088	0.581	0.284	0.101
Leverage	Pressure drop	90.074	10.117	9.408	0.551	0.450	0.088
oracle	Cooling objective	90.000	2.695	1.394	1.000	1.000	1.000
oracle	Pressure drop	90.000	8.047	4.098	1.000	1.000	1.000
Random	Cooling objective	89.330	3.510	3.488	0.501	0.228	-0.007
Random	Pressure drop	89.330	10.347	10.355	0.500	0.413	-0.006
Coordinate gap	Cooling objective	88.696	3.497	3.536	0.484	0.222	-0.013
Coordinate gap	Pressure drop	88.696	10.361	10.332	0.498	0.403	0.006

Table 10: XGB score ablation on design-random tests. AURC is tie-aware and undefined for constant scores; it is a ranking diagnostic, not a risk guarantee.

intervals

Model	Target	Designs	Coverage band %	Design risk band	Strata	Stratum risk band
ET-seed	Cooling objective	1790	[88.3, 91.1]	[4.380, 4.764]	6	[4.350, 4.868]
ET-seed	Pressure drop	1790	[89.6, 92.0]	[13.171, 14.252]	6	[13.127, 14.388]
XGB	Cooling objective	1790	[88.4, 91.1]	[3.218, 3.546]	6	[3.165, 3.618]
XGB	Pressure drop	1790	[88.3, 91.0]	[9.443, 10.239]	6	[9.400, 10.170]
GP-ARD	Cooling objective	1790	[89.3, 91.8]	[2.113, 2.271]	6	[2.144, 2.244]
GP-ARD	Pressure drop	1790	[89.7, 92.5]	[7.014, 7.641]	6	[7.031, 7.599]

Table 11: Seed 42; 500 resamples conditional on fitted model. One row per unique design. Six covariate-only KMeans strata give coarse descriptive bands, not confidence intervals for unseen physical geometry families.

competitors

Model	Target	Score	Coverage %	Ret. MAPE %	Interval coverage %
CQR	Cooling objective	CQR width	89.348	3.215	89.609
CQR	Cooling objective	Uncertainty	89.181	3.230	89.609
CQR	Cooling objective	Max U+D	88.994	3.284	89.609
CQR	Pressure drop	CQR width	90.521	10.151	88.901
CQR	Pressure drop	Uncertainty	90.912	9.865	88.901
CQR	Pressure drop	Max U+D	90.149	10.108	88.901
ET-bag	Cooling objective	Uncertainty	88.231	4.862	–
ET-bag	Cooling objective	Max U+D	89.423	4.805	–
ET-bag	Pressure drop	Uncertainty	89.683	13.824	–
ET-bag	Pressure drop	Max U+D	89.739	13.976	–
ET-seed	Cooling objective	Uncertainty	89.777	4.795	–
ET-seed	Cooling objective	Max U+D	89.050	4.713	–
ET-seed	Pressure drop	Uncertainty	89.683	13.069	–
ET-seed	Pressure drop	Max U+D	89.609	13.327	–
GP-ARD	Cooling objective	Uncertainty	89.255	2.188	–
GP-ARD	Cooling objective	Max U+D	89.851	2.204	–
GP-ARD	Pressure drop	Uncertainty	89.832	7.250	–
GP-ARD	Pressure drop	Max U+D	89.777	7.292	–
Log-poly	Cooling objective	Uncertainty	100.000	2.554	–
Log-poly	Cooling objective	Max U+D	90.354	2.480	–
Log-poly	Pressure drop	Uncertainty	100.000	10.481	–
Log-poly	Pressure drop	Max U+D	90.354	10.173	–
XGB	Cooling objective	Uncertainty	89.330	3.352	–
XGB	Cooling objective	Max U+D	89.460	3.360	–
XGB	Pressure drop	Uncertainty	89.404	9.660	–
XGB	Pressure drop	Max U+D	89.181	9.750	–

Table 12: Point-selection risk and CQR interval coverage are different estimands. GP pressure point estimates are inverse-log medians, while tree ensemble estimates are physical-unit member means.

matched risk

Model	Target	Distance	Uncertainty	Equal U+D	Max U+D
CQR	Cooling objective	3.404	3.240	3.254	3.295
CQR	Pressure drop	10.606	9.849	9.927	10.091
ET-bag	Cooling objective	4.840	4.874	4.816	4.813
ET-bag	Pressure drop	14.300	13.822	13.888	13.983
ET-seed	Cooling objective	4.729	4.792	4.710	4.729
ET-seed	Pressure drop	13.598	13.110	13.130	13.360
GP-ARD	Cooling objective	2.231	2.191	2.199	2.205
GP-ARD	Pressure drop	7.313	7.260	7.272	7.280
Log-poly	Cooling objective	2.483	2.554	2.483	2.483
Log-poly	Pressure drop	10.148	10.481	10.148	10.148
XGB	Cooling objective	3.399	3.360	3.333	3.369
XGB	Pressure drop	10.150	9.664	9.733	9.781

Table 13: MAPE at approximately 90% expected retained coverage, averaged over three runs. Random ordering inside ties may be unattainable by one deterministic cutoff.

binary sweep

Table 14: XGB seed-42 sensitivity. Binary calibration quantiles vary the cutoff; tolerance sweeps vary failure labels without refitting or moving the cutoff.

Fold	Target	calibration_quantile	Coverage %	Ret. MAPE %	Ret. fail %
Joint high	Cooling objective	0.850	37.845	4.756	40.876
Joint high	Cooling objective	0.900	48.619	5.027	41.477
Joint high	Cooling objective	0.950	63.536	4.968	41.304
Joint high	Cooling objective	0.980	78.729	4.982	42.807
Joint high	Pressure drop	0.850	37.845	9.638	36.496
Joint high	Pressure drop	0.900	46.133	9.746	38.323
Joint high	Pressure drop	0.950	64.917	10.049	39.574
Joint high	Pressure drop	0.980	80.110	10.392	40.690
Joint low	Cooling objective	0.850	40.164	4.906	37.415
Joint low	Cooling objective	0.900	51.913	4.954	36.842
Joint low	Cooling objective	0.950	66.940	5.039	37.143
Joint low	Cooling objective	0.980	83.060	5.069	38.158
Joint low	Pressure drop	0.850	44.809	23.563	89.024
Joint low	Pressure drop	0.900	55.464	22.768	84.729
Joint low	Pressure drop	0.950	65.847	23.290	83.817
Joint low	Pressure drop	0.980	84.973	23.583	82.958
Design random	Cooling objective	0.850	84.637	3.338	21.452
Design random	Cooling objective	0.900	89.777	3.391	21.780
Design random	Cooling objective	0.950	94.581	3.440	22.209
Design random	Cooling objective	0.980	98.771	3.476	22.738
Design random	Pressure drop	0.850	86.760	9.735	38.957
Design random	Pressure drop	0.900	89.777	9.832	39.203
Design random	Pressure drop	0.950	96.034	10.047	40.140
Design random	Pressure drop	0.980	98.268	10.265	40.762
x0 high	Cooling objective	0.850	69.609	3.027	19.101
x0 high	Cooling objective	0.900	79.441	3.112	19.831
x0 high	Cooling objective	0.950	88.324	3.170	20.051
x0 high	Cooling objective	0.980	94.246	3.221	20.569
x0 high	Pressure drop	0.850	69.162	8.506	32.956
x0 high	Pressure drop	0.900	79.274	8.697	33.474
x0 high	Pressure drop	0.950	89.665	8.949	34.206
x0 high	Pressure drop	0.980	96.369	9.043	34.899
x0 low	Cooling objective	0.850	65.866	5.829	47.074
x0 low	Cooling objective	0.900	71.788	5.929	47.471
x0 low	Cooling objective	0.950	86.592	6.096	48.968
x0 low	Cooling objective	0.980	94.358	6.186	49.497
x0 low	Pressure drop	0.850	66.201	17.474	66.751
x0 low	Pressure drop	0.900	75.196	17.751	67.088
x0 low	Pressure drop	0.950	89.162	18.179	68.296

Fold	Target	calibration_quantile	Coverage %	Ret. MAPE %	Ret. fail %
x0 low	Pressure drop	0.980	95.587	18.307	68.381
x0 middle	Cooling objective	0.850	88.492	3.512	23.927
x0 middle	Cooling objective	0.900	92.737	3.520	23.976
x0 middle	Cooling objective	0.950	96.313	3.570	24.420
x0 middle	Cooling objective	0.980	98.436	3.619	24.972
x0 middle	Pressure drop	0.850	88.212	10.433	42.875
x0 middle	Pressure drop	0.900	91.620	10.465	42.988
x0 middle	Pressure drop	0.950	96.257	10.644	43.471
x0 middle	Pressure drop	0.980	98.603	10.806	44.023
x12 high	Cooling objective	0.850	69.050	3.294	20.550
x12 high	Cooling objective	0.900	75.475	3.398	21.392
x12 high	Cooling objective	0.950	85.642	3.530	22.896
x12 high	Cooling objective	0.980	91.508	3.562	23.382
x12 high	Pressure drop	0.850	70.112	9.179	36.096
x12 high	Pressure drop	0.900	79.050	9.568	37.597
x12 high	Pressure drop	0.950	86.704	9.815	38.402
x12 high	Pressure drop	0.980	92.235	9.836	38.583
x12 low	Cooling objective	0.850	66.983	3.433	22.602
x12 low	Cooling objective	0.900	75.587	3.523	23.429
x12 low	Cooling objective	0.950	86.592	3.524	23.742
x12 low	Cooling objective	0.980	93.855	3.533	23.690
x12 low	Pressure drop	0.850	66.983	10.095	40.784
x12 low	Pressure drop	0.900	74.916	10.197	41.909
x12 low	Pressure drop	0.950	87.598	10.240	41.773
x12 low	Pressure drop	0.980	95.978	10.352	42.433
x12 middle	Cooling objective	0.850	88.212	3.403	21.913
x12 middle	Cooling objective	0.900	91.285	3.409	22.032
x12 middle	Cooling objective	0.950	95.978	3.458	22.293
x12 middle	Cooling objective	0.980	98.715	3.496	22.807
x12 middle	Pressure drop	0.850	87.486	9.975	39.719
x12 middle	Pressure drop	0.900	91.564	10.142	40.513
x12 middle	Pressure drop	0.950	97.263	10.332	41.126
x12 middle	Pressure drop	0.980	98.436	10.369	41.203
x6 high	Cooling objective	0.850	67.151	4.011	30.200
x6 high	Cooling objective	0.900	74.749	4.134	31.764
x6 high	Cooling objective	0.950	86.983	4.237	32.627
x6 high	Cooling objective	0.980	93.017	4.296	33.213
x6 high	Pressure drop	0.850	67.039	10.868	44.250
x6 high	Pressure drop	0.900	74.190	10.986	44.202
x6 high	Pressure drop	0.950	86.648	11.280	44.939
x6 high	Pressure drop	0.980	93.799	11.384	45.205
x6 low	Cooling objective	0.850	65.475	4.678	41.041
x6 low	Cooling objective	0.900	74.302	4.679	40.677
x6 low	Cooling objective	0.950	85.028	4.722	41.261
x6 low	Cooling objective	0.980	88.994	4.721	41.243
x6 low	Pressure drop	0.850	67.877	9.183	37.037

Fold	Target	calibration_quantile	Coverage %	Ret. MAPE %	Ret. fail %
x6 low	Pressure drop	0.900	76.201	9.419	37.463
x6 low	Pressure drop	0.950	86.480	9.715	38.760
x6 low	Pressure drop	0.980	92.067	9.783	39.017
x6 middle	Cooling objective	0.850	88.771	3.696	23.789
x6 middle	Cooling objective	0.900	92.235	3.702	23.925
x6 middle	Cooling objective	0.950	95.642	3.719	24.241
x6 middle	Cooling objective	0.980	97.709	3.728	24.357
x6 middle	Pressure drop	0.850	89.721	9.748	37.422
x6 middle	Pressure drop	0.900	92.961	9.846	37.861
x6 middle	Pressure drop	0.950	96.089	9.991	38.198
x6 middle	Pressure drop	0.980	98.101	10.033	38.497
Supplied random	Cooling objective	0.850	84.022	3.206	20.346
Supplied random	Cooling objective	0.900	89.832	3.205	20.522
Supplied random	Cooling objective	0.950	94.302	3.221	20.853
Supplied random	Cooling objective	0.980	97.765	3.245	21.029
Supplied random	Pressure drop	0.850	86.257	9.208	36.399
Supplied random	Pressure drop	0.900	90.391	9.286	37.083
Supplied random	Pressure drop	0.950	94.413	9.376	37.396
Supplied random	Pressure drop	0.980	97.207	9.499	38.161

tolerances

Table 15: XGB seed-42 sensitivity. Binary calibration quantiles vary the cutoff; tolerance sweeps vary failure labels without refitting or moving the cutoff.

Fold	Target	tolerance	Coverage %	Ret. MAPE %	Ret. fail %
Joint high	Cooling objective	1	48.619	5.027	86.932
Joint high	Cooling objective	2	48.619	5.027	76.705
Joint high	Cooling objective	5	48.619	5.027	41.477
Joint high	Cooling objective	10	48.619	5.027	11.932
Joint high	Cooling objective	15	48.619	5.027	1.136
Joint high	Pressure drop	2	46.133	9.746	88.024
Joint high	Pressure drop	5	46.133	9.746	71.257
Joint high	Pressure drop	10	46.133	9.746	38.323
Joint high	Pressure drop	15	46.133	9.746	20.958
Joint high	Pressure drop	20	46.133	9.746	10.180
Joint low	Cooling objective	1	51.913	4.954	88.421
Joint low	Cooling objective	2	51.913	4.954	72.105
Joint low	Cooling objective	5	51.913	4.954	36.842
Joint low	Cooling objective	10	51.913	4.954	10.526
Joint low	Cooling objective	15	51.913	4.954	3.158
Joint low	Pressure drop	2	55.464	22.768	98.030
Joint low	Pressure drop	5	55.464	22.768	93.103
Joint low	Pressure drop	10	55.464	22.768	84.729
Joint low	Pressure drop	15	55.464	22.768	72.414
Joint low	Pressure drop	20	55.464	22.768	57.635
Design random	Cooling objective	1	89.777	3.391	79.714
Design random	Cooling objective	2	89.777	3.391	60.921
Design random	Cooling objective	5	89.777	3.391	21.780
Design random	Cooling objective	10	89.777	3.391	3.485
Design random	Cooling objective	15	89.777	3.391	1.182
Design random	Pressure drop	2	89.777	9.832	85.625
Design random	Pressure drop	5	89.777	9.832	66.024
Design random	Pressure drop	10	89.777	9.832	39.203
Design random	Pressure drop	15	89.777	9.832	20.971
Design random	Pressure drop	20	89.777	9.832	11.325
x0 high	Cooling objective	1	79.441	3.112	77.286
x0 high	Cooling objective	2	79.441	3.112	57.454
x0 high	Cooling objective	5	79.441	3.112	19.831
x0 high	Cooling objective	10	79.441	3.112	2.250
x0 high	Cooling objective	15	79.441	3.112	0.422
x0 high	Pressure drop	2	79.274	8.697	84.637
x0 high	Pressure drop	5	79.274	8.697	60.606
x0 high	Pressure drop	10	79.274	8.697	33.474
x0 high	Pressure drop	15	79.274	8.697	16.843
x0 high	Pressure drop	20	79.274	8.697	8.739

Fold	Target	tolerance	Coverage %	Ret. MAPE %	Ret. fail %
x0 low	Cooling objective	1	71.788	5.929	90.195
x0 low	Cooling objective	2	71.788	5.929	80.545
x0 low	Cooling objective	5	71.788	5.929	47.471
x0 low	Cooling objective	10	71.788	5.929	17.432
x0 low	Cooling objective	15	71.788	5.929	5.058
x0 low	Pressure drop	2	75.196	17.751	93.239
x0 low	Pressure drop	5	75.196	17.751	82.392
x0 low	Pressure drop	10	75.196	17.751	67.088
x0 low	Pressure drop	15	75.196	17.751	53.418
x0 low	Pressure drop	20	75.196	17.751	40.416
x0 middle	Cooling objective	1	92.737	3.520	82.590
x0 middle	Cooling objective	2	92.737	3.520	65.482
x0 middle	Cooling objective	5	92.737	3.520	23.976
x0 middle	Cooling objective	10	92.737	3.520	2.831
x0 middle	Cooling objective	15	92.737	3.520	0.723
x0 middle	Pressure drop	2	91.620	10.465	87.500
x0 middle	Pressure drop	5	91.620	10.465	69.207
x0 middle	Pressure drop	10	91.620	10.465	42.988
x0 middle	Pressure drop	15	91.620	10.465	24.390
x0 middle	Pressure drop	20	91.620	10.465	13.110
x12 high	Cooling objective	1	75.475	3.398	81.421
x12 high	Cooling objective	2	75.475	3.398	62.990
x12 high	Cooling objective	5	75.475	3.398	21.392
x12 high	Cooling objective	10	75.475	3.398	2.887
x12 high	Cooling objective	15	75.475	3.398	0.962
x12 high	Pressure drop	2	79.050	9.568	87.491
x12 high	Pressure drop	5	79.050	9.568	64.382
x12 high	Pressure drop	10	79.050	9.568	37.597
x12 high	Pressure drop	15	79.050	9.568	20.636
x12 high	Pressure drop	20	79.050	9.568	10.459
x12 low	Cooling objective	1	75.587	3.523	80.266
x12 low	Cooling objective	2	75.587	3.523	61.493
x12 low	Cooling objective	5	75.587	3.523	23.429
x12 low	Cooling objective	10	75.587	3.523	3.991
x12 low	Cooling objective	15	75.587	3.523	0.961
x12 low	Pressure drop	2	74.916	10.197	86.875
x12 low	Pressure drop	5	74.916	10.197	68.680
x12 low	Pressure drop	10	74.916	10.197	41.909
x12 low	Pressure drop	15	74.916	10.197	21.924
x12 low	Pressure drop	20	74.916	10.197	12.528
x12 middle	Cooling objective	1	91.285	3.409	80.233
x12 middle	Cooling objective	2	91.285	3.409	62.301
x12 middle	Cooling objective	5	91.285	3.409	22.032
x12 middle	Cooling objective	10	91.285	3.409	3.121
x12 middle	Cooling objective	15	91.285	3.409	0.857
x12 middle	Pressure drop	2	91.564	10.142	86.455

Fold	Target	tolerance	Coverage %	Ret. MAPE %	Ret. fail %
x12 middle	Pressure drop	5	91.564	10.142	66.382
x12 middle	Pressure drop	10	91.564	10.142	40.513
x12 middle	Pressure drop	15	91.564	10.142	22.819
x12 middle	Pressure drop	20	91.564	10.142	12.569
x6 high	Cooling objective	1	74.749	4.134	81.839
x6 high	Cooling objective	2	74.749	4.134	66.891
x6 high	Cooling objective	5	74.749	4.134	31.764
x6 high	Cooling objective	10	74.749	4.134	6.577
x6 high	Cooling objective	15	74.749	4.134	1.420
x6 high	Pressure drop	2	74.190	10.986	86.973
x6 high	Pressure drop	5	74.190	10.986	69.654
x6 high	Pressure drop	10	74.190	10.986	44.202
x6 high	Pressure drop	15	74.190	10.986	26.506
x6 high	Pressure drop	20	74.190	10.986	14.157
x6 low	Cooling objective	1	74.302	4.679	89.925
x6 low	Cooling objective	2	74.302	4.679	79.474
x6 low	Cooling objective	5	74.302	4.679	40.677
x6 low	Cooling objective	10	74.302	4.679	4.737
x6 low	Cooling objective	15	74.302	4.679	0.301
x6 low	Pressure drop	2	76.201	9.419	84.824
x6 low	Pressure drop	5	76.201	9.419	64.589
x6 low	Pressure drop	10	76.201	9.419	37.463
x6 low	Pressure drop	15	76.201	9.419	20.015
x6 low	Pressure drop	20	76.201	9.419	10.191
x6 middle	Cooling objective	1	92.235	3.702	82.374
x6 middle	Cooling objective	2	92.235	3.702	64.204
x6 middle	Cooling objective	5	92.235	3.702	23.925
x6 middle	Cooling objective	10	92.235	3.702	4.603
x6 middle	Cooling objective	15	92.235	3.702	1.878
x6 middle	Pressure drop	2	92.961	9.846	85.938
x6 middle	Pressure drop	5	92.961	9.846	64.904
x6 middle	Pressure drop	10	92.961	9.846	37.861
x6 middle	Pressure drop	15	92.961	9.846	21.214
x6 middle	Pressure drop	20	92.961	9.846	11.298
Supplied random	Cooling objective	1	89.832	3.205	81.841
Supplied random	Cooling objective	2	89.832	3.205	61.443
Supplied random	Cooling objective	5	89.832	3.205	20.522
Supplied random	Cooling objective	10	89.832	3.205	1.741
Supplied random	Cooling objective	15	89.832	3.205	0.249
Supplied random	Pressure drop	2	90.391	9.286	85.661
Supplied random	Pressure drop	5	90.391	9.286	64.400
Supplied random	Pressure drop	10	90.391	9.286	37.083
Supplied random	Pressure drop	15	90.391	9.286	18.047
Supplied random	Pressure drop	20	90.391	9.286	9.889

conformal groups

Fold	Target	Row coverage %	Design coverage %	Mean width
Joint high	Cooling objective	87.109	87.109	1.675
Joint high	Pressure drop	94.199	94.199	40.751
Joint low	Cooling objective	91.985	91.985	1.761
Joint low	Pressure drop	39.344	39.344	38.666
Design random	Cooling objective	89.609	89.609	1.579
Design random	Pressure drop	88.901	88.901	36.990
x0 high	Cooling objective	93.315	93.315	1.686
x0 high	Pressure drop	94.432	94.432	38.285
x0 low	Cooling objective	81.676	81.676	1.618
x0 low	Pressure drop	66.294	66.294	40.303
x0 middle	Cooling objective	88.026	88.026	1.641
x0 middle	Pressure drop	88.939	88.939	36.846
x12 high	Cooling objective	90.261	90.261	1.680
x12 high	Pressure drop	90.782	90.782	38.937
x12 low	Cooling objective	88.901	88.901	1.619
x12 low	Pressure drop	90.112	90.112	39.606
x12 middle	Cooling objective	91.620	91.620	1.666
x12 middle	Pressure drop	90.931	90.931	39.773
x6 high	Cooling objective	90.298	90.298	1.721
x6 high	Pressure drop	89.330	89.330	39.587
x6 low	Cooling objective	77.970	77.970	1.767
x6 low	Pressure drop	87.747	87.747	37.943
x6 middle	Cooling objective	88.585	88.585	1.548
x6 middle	Pressure drop	90.596	90.596	37.661
Supplied random	Cooling objective	91.173	91.173	1.552
Supplied random	Pressure drop	90.354	90.354	37.610

Table 16: CQR 90% intervals, three-run means. Each design contains one row, so row/design coverage coincide. Group exchangeability is hypothetical under Sobol sampling and does not hold automatically for imposed support shifts. Width is in physical target units.

point baselines

Fold	Target	Model	Fit rows	MAPE %	RMSE	R2
Design random	Cooling objective	ExtraTrees	7156.000	4.712	0.412	0.812
Design random	Cooling objective	HistGradientBoosting	7156.000	2.962	0.262	0.924
Design random	Cooling objective	RandomForest	7156.000	4.787	0.420	0.804
Design random	Cooling objective	SVR_RBF	7156.000	2.447	0.216	0.948
Design random	Cooling objective	XGBoost	7156.000	3.035	0.271	0.919
Design random	Pressure drop	ExtraTrees	7156.000	13.030	21.492	0.569
Design random	Pressure drop	HistGradientBoosting	7156.000	9.016	14.796	0.796
Design random	Pressure drop	RandomForest	7156.000	13.466	22.042	0.546
Design random	Pressure drop	SVR_RBF	7156.000	8.677	14.816	0.794
Design random	Pressure drop	XGBoost	7156.000	9.207	15.467	0.777
Supplied random	Cooling objective	ExtraTrees	8051.000	4.481	0.388	0.833
Supplied random	Cooling objective	HistGradientBoosting	8051.000	2.791	0.237	0.938
Supplied random	Cooling objective	RandomForest	8051.000	4.599	0.394	0.828
Supplied random	Cooling objective	SVR_RBF	8051.000	2.335	0.203	0.954
Supplied random	Cooling objective	XGBoost	8051.000	2.872	0.246	0.933
Supplied random	Pressure drop	ExtraTrees	8051.000	12.524	25.397	0.519
Supplied random	Pressure drop	HistGradientBoosting	8051.000	8.359	17.837	0.763
Supplied random	Pressure drop	RandomForest	8051.000	12.697	24.785	0.542
Supplied random	Pressure drop	SVR_RBF	8051.000	8.300	20.204	0.695
Supplied random	Pressure drop	XGBoost	8051.000	8.437	17.621	0.768

Table 17: Original five point-regressor families on full outer training. These are context benchmarks, not training-budget-matched selective models; RMSE is in target units.

fit budget

Fold	Target	Model	Fit budget	Fit rows	MAPE %
Design random	Cooling objective	ExtraTrees_single	fit	4294.000	4.874
Design random	Cooling objective	ExtraTrees_single	outer_training	7156.000	4.731
Design random	Cooling objective	SVR_tuned	fit	4294.000	2.372
Design random	Pressure drop	ExtraTrees_single	fit	4294.000	13.816
Design random	Pressure drop	ExtraTrees_single	outer_training	7156.000	13.089
Design random	Pressure drop	SVR_tuned	fit	4294.000	9.141
Supplied random	Cooling objective	ExtraTrees_single	fit	4830.000	4.644
Supplied random	Cooling objective	ExtraTrees_single	outer_training	8051.000	4.493
Supplied random	Cooling objective	SVR_tuned	fit	4830.000	2.298
Supplied random	Pressure drop	ExtraTrees_single	fit	4830.000	13.272
Supplied random	Pressure drop	ExtraTrees_single	outer_training	8051.000	12.552
Supplied random	Pressure drop	SVR_tuned	fit	4830.000	8.647

Table 18: Single ET on common fit versus full outer train and tune-selected converged SVR on common fit. Three-run means isolate training budget from selective scoring.

sensitivity

Lower q	Upper q	Lower cutoff	Upper cutoff	Predict %	Warn %	Predict MAPE %	Warn MAPE %
0.600	0.850	0.681	0.937	57.318	27.318	3.111	3.813
0.600	0.900	0.681	1.026	57.318	32.458	3.111	3.884
0.600	0.950	0.681	1.145	57.318	37.263	3.111	3.947
0.600	0.980	0.681	1.325	57.318	41.453	3.111	3.980
0.750	0.850	0.820	0.937	72.905	11.732	3.207	4.150
0.750	0.900	0.820	1.026	72.905	16.872	3.207	4.184
0.750	0.950	0.820	1.145	72.905	21.676	3.207	4.226
0.750	0.980	0.820	1.325	72.905	25.866	3.207	4.234
0.800	0.850	0.870	0.937	78.547	6.089	3.251	4.459
0.800	0.900	0.870	1.026	78.547	11.229	3.251	4.369
0.800	0.950	0.870	1.145	78.547	16.034	3.251	4.370
0.800	0.980	0.870	1.325	78.547	20.223	3.251	4.350
0.850	0.900	0.937	1.026	84.637	5.140	3.338	4.262
0.850	0.950	0.937	1.145	84.637	9.944	3.338	4.315
0.850	0.980	0.937	1.325	84.637	14.134	3.338	4.303

Table 19: Three-state diagnostic for XGB cooling objective, seed 42/design-random. Primary decisions are binary; no intermediate risk ordering is presumed.

actual thresholds

Table 20: Independent 90th-percentile binary calibration cutoffs and actual test retention, seed 42. Unbounded scores permit cutoffs above one; cutoffs are not error-probability thresholds.

Fold	Target	Model	Cal. cutoff	Test coverage %	Ret. MAPE %
Joint high	Cooling objective	CQR	0.984	49.448	4.830
Joint high	Cooling objective	ET-bag	0.991	48.343	5.865
Joint high	Cooling objective	ET-seed	1.004	53.591	5.854
Joint high	Cooling objective	GP-ARD	0.941	17.680	2.891
Joint high	Cooling objective	Log-poly	0.899	38.950	3.322
Joint high	Cooling objective	XGB	0.975	48.619	5.027
Joint high	Pressure drop	CQR	0.993	48.066	9.921
Joint high	Pressure drop	ET-bag	0.924	40.884	13.141
Joint high	Pressure drop	ET-seed	0.964	45.304	11.959
Joint high	Pressure drop	GP-ARD	0.973	34.807	10.673
Joint high	Pressure drop	Log-poly	0.899	38.950	15.433
Joint high	Pressure drop	XGB	0.951	46.133	9.746
Joint low	Cooling objective	CQR	1.026	51.639	4.972
Joint low	Cooling objective	ET-bag	1.008	49.727	5.806
Joint low	Cooling objective	ET-seed	1.026	53.279	5.716
Joint low	Cooling objective	GP-ARD	0.980	39.071	2.866
Joint low	Cooling objective	Log-poly	0.955	46.448	4.038
Joint low	Cooling objective	XGB	1.021	51.913	4.954
Joint low	Pressure drop	CQR	1.084	59.563	23.626
Joint low	Pressure drop	ET-bag	1.039	55.738	27.364
Joint low	Pressure drop	ET-seed	1.057	58.197	27.361
Joint low	Pressure drop	GP-ARD	0.997	30.874	10.463
Joint low	Pressure drop	Log-poly	0.955	46.448	14.164
Joint low	Pressure drop	XGB	1.057	55.464	22.768
Design random	Cooling objective	CQR	1.030	90.056	3.340
Design random	Cooling objective	ET-bag	1.030	90.112	4.683
Design random	Cooling objective	ET-seed	1.029	89.721	4.576
Design random	Cooling objective	GP-ARD	0.995	90.447	2.192
Design random	Cooling objective	Log-poly	0.971	92.179	2.494
Design random	Cooling objective	XGB	1.026	89.777	3.391
Design random	Pressure drop	CQR	1.045	91.061	10.336
Design random	Pressure drop	ET-bag	1.038	89.665	13.961
Design random	Pressure drop	ET-seed	1.038	90.782	13.694
Design random	Pressure drop	GP-ARD	1.017	91.061	7.327
Design random	Pressure drop	Log-poly	0.971	92.179	10.069
Design random	Pressure drop	XGB	1.062	89.777	9.832
x0 high	Cooling objective	CQR	1.042	76.760	3.103
x0 high	Cooling objective	ET-bag	1.048	76.201	4.191
x0 high	Cooling objective	ET-seed	1.048	77.989	4.152

Fold	Target	Model	Cal. cutoff	Test coverage %	Ret. MAPE %
x0 high	Cooling objective	GP-ARD	1.026	66.480	2.590
x0 high	Cooling objective	Log-poly	0.968	70.615	2.745
x0 high	Cooling objective	XGB	1.070	79.441	3.112
x0 high	Pressure drop	CQR	1.062	77.654	8.912
x0 high	Pressure drop	ET-bag	1.064	78.212	11.249
x0 high	Pressure drop	ET-seed	1.102	81.006	10.822
x0 high	Pressure drop	GP-ARD	1.015	54.302	8.392
x0 high	Pressure drop	Log-poly	0.968	70.615	13.264
x0 high	Pressure drop	XGB	1.062	79.274	8.697
x0 low	Cooling objective	CQR	0.978	72.682	5.889
x0 low	Cooling objective	ET-bag	1.008	75.754	7.331
x0 low	Cooling objective	ET-seed	1.029	77.095	7.406
x0 low	Cooling objective	GP-ARD	0.975	51.397	2.997
x0 low	Cooling objective	Log-poly	0.922	70.670	3.462
x0 low	Cooling objective	XGB	0.984	71.788	5.929
x0 low	Pressure drop	CQR	0.996	72.346	17.535
x0 low	Pressure drop	ET-bag	1.015	75.978	20.993
x0 low	Pressure drop	ET-seed	1.014	76.872	21.093
x0 low	Pressure drop	GP-ARD	0.946	57.877	9.904
x0 low	Pressure drop	Log-poly	0.922	70.670	13.851
x0 low	Pressure drop	XGB	1.003	75.196	17.751
x0 middle	Cooling objective	CQR	1.031	91.564	3.404
x0 middle	Cooling objective	ET-bag	1.019	90.894	5.015
x0 middle	Cooling objective	ET-seed	1.024	91.173	4.893
x0 middle	Cooling objective	GP-ARD	1.019	93.855	2.201
x0 middle	Cooling objective	Log-poly	0.956	92.682	2.432
x0 middle	Cooling objective	XGB	1.047	92.737	3.520
x0 middle	Pressure drop	CQR	1.031	92.346	10.210
x0 middle	Pressure drop	ET-bag	1.042	91.899	15.859
x0 middle	Pressure drop	ET-seed	1.034	90.279	14.856
x0 middle	Pressure drop	GP-ARD	1.018	93.296	7.676
x0 middle	Pressure drop	Log-poly	0.956	92.682	9.751
x0 middle	Pressure drop	XGB	1.034	91.620	10.465
x12 high	Cooling objective	CQR	1.010	75.978	3.378
x12 high	Cooling objective	ET-bag	1.059	78.994	4.782
x12 high	Cooling objective	ET-seed	1.060	80.056	4.725
x12 high	Cooling objective	GP-ARD	1.016	77.263	2.388
x12 high	Cooling objective	Log-poly	0.935	71.620	2.504
x12 high	Cooling objective	XGB	1.027	75.475	3.398
x12 high	Pressure drop	CQR	1.018	76.872	9.542
x12 high	Pressure drop	ET-bag	1.054	78.659	12.607
x12 high	Pressure drop	ET-seed	1.037	78.045	12.133
x12 high	Pressure drop	GP-ARD	1.003	77.709	6.778
x12 high	Pressure drop	Log-poly	0.935	71.620	10.413
x12 high	Pressure drop	XGB	1.059	79.050	9.568
x12 low	Cooling objective	CQR	1.050	75.978	3.420

Fold	Target	Model	Cal. cutoff	Test coverage %	Ret. MAPE %
x12 low	Cooling objective	ET-bag	1.044	75.196	4.867
x12 low	Cooling objective	ET-seed	1.033	73.743	4.708
x12 low	Cooling objective	GP-ARD	1.020	74.246	2.446
x12 low	Cooling objective	Log-poly	0.968	70.335	2.528
x12 low	Cooling objective	XGB	1.061	75.587	3.523
x12 low	Pressure drop	CQR	1.009	71.173	10.226
x12 low	Pressure drop	ET-bag	1.028	73.184	14.624
x12 low	Pressure drop	ET-seed	1.053	74.916	13.796
x12 low	Pressure drop	GP-ARD	1.022	75.028	7.475
x12 low	Pressure drop	Log-poly	0.968	70.335	9.995
x12 low	Pressure drop	XGB	1.050	74.916	10.197
x12 middle	Cooling objective	CQR	1.018	91.173	3.279
x12 middle	Cooling objective	ET-bag	0.997	90.447	4.741
x12 middle	Cooling objective	ET-seed	1.003	91.676	4.664
x12 middle	Cooling objective	GP-ARD	1.000	91.732	2.201
x12 middle	Cooling objective	Log-poly	0.944	93.184	2.434
x12 middle	Cooling objective	XGB	0.997	91.285	3.409
x12 middle	Pressure drop	CQR	1.011	91.229	10.168
x12 middle	Pressure drop	ET-bag	1.009	91.229	14.285
x12 middle	Pressure drop	ET-seed	1.009	91.117	13.629
x12 middle	Pressure drop	GP-ARD	0.998	91.173	7.248
x12 middle	Pressure drop	Log-poly	0.944	93.184	10.257
x12 middle	Pressure drop	XGB	1.031	91.564	10.142
x6 high	Cooling objective	CQR	1.016	73.296	3.915
x6 high	Cooling objective	ET-bag	1.010	72.905	5.471
x6 high	Cooling objective	ET-seed	1.019	73.520	5.183
x6 high	Cooling objective	GP-ARD	1.010	69.385	2.431
x6 high	Cooling objective	Log-poly	0.951	70.112	2.705
x6 high	Cooling objective	XGB	1.042	74.749	4.134
x6 high	Pressure drop	CQR	1.050	74.804	11.321
x6 high	Pressure drop	ET-bag	1.048	76.201	16.788
x6 high	Pressure drop	ET-seed	1.016	73.631	15.945
x6 high	Pressure drop	GP-ARD	0.998	74.078	7.064
x6 high	Pressure drop	Log-poly	0.951	70.112	10.843
x6 high	Pressure drop	XGB	1.016	74.190	10.986
x6 low	Cooling objective	CQR	1.049	76.145	4.668
x6 low	Cooling objective	ET-bag	1.052	73.743	6.700
x6 low	Cooling objective	ET-seed	1.025	75.363	6.408
x6 low	Cooling objective	GP-ARD	1.022	75.978	2.326
x6 low	Cooling objective	Log-poly	0.965	71.844	2.439
x6 low	Cooling objective	XGB	1.029	74.302	4.679
x6 low	Pressure drop	CQR	1.042	77.709	9.984
x6 low	Pressure drop	ET-bag	1.035	75.754	12.513
x6 low	Pressure drop	ET-seed	1.053	78.492	12.013
x6 low	Pressure drop	GP-ARD	1.023	75.866	7.593
x6 low	Pressure drop	Log-poly	0.965	71.844	9.764

Fold	Target	Model	Cal. cutoff	Test coverage %	Ret. MAPE %
x6 low	Pressure drop	XGB	1.046	76.201	9.419
x6 middle	Cooling objective	CQR	1.108	95.363	3.451
x6 middle	Cooling objective	ET-bag	1.091	96.536	4.891
x6 middle	Cooling objective	ET-seed	1.095	92.514	4.677
x6 middle	Cooling objective	GP-ARD	1.091	96.201	2.313
x6 middle	Cooling objective	Log-poly	1.005	94.525	2.490
x6 middle	Cooling objective	XGB	1.060	92.235	3.702
x6 middle	Pressure drop	CQR	1.099	94.469	10.113
x6 middle	Pressure drop	ET-bag	1.090	93.575	14.106
x6 middle	Pressure drop	ET-seed	1.048	91.397	13.105
x6 middle	Pressure drop	GP-ARD	1.062	93.631	7.583
x6 middle	Pressure drop	Log-poly	1.005	94.525	10.260
x6 middle	Pressure drop	XGB	1.052	92.961	9.846
Supplied random	Cooling objective	CQR	1.053	89.944	3.154
Supplied random	Cooling objective	ET-bag	1.044	90.279	4.702
Supplied random	Cooling objective	ET-seed	1.053	89.609	4.531
Supplied random	Cooling objective	GP-ARD	1.028	90.950	2.218
Supplied random	Cooling objective	Log-poly	0.981	90.726	2.467
Supplied random	Cooling objective	XGB	1.034	89.832	3.205
Supplied random	Pressure drop	CQR	1.043	89.832	9.641
Supplied random	Pressure drop	ET-bag	1.029	89.832	13.458
Supplied random	Pressure drop	ET-seed	1.044	89.609	12.747
Supplied random	Pressure drop	GP-ARD	1.021	90.615	6.989
Supplied random	Pressure drop	Log-poly	0.981	90.726	9.800
Supplied random	Pressure drop	XGB	1.045	90.391	9.286

error distribution

Table 21: Error-distribution summaries averaged over three runs. Symmetric APE is bounded by 200%; raw per-run quantiles, maxima and numerical guard counts are exported separately.

Fold	Target	Model	Median APE %	95th APE %	Mean sMAPE %	Nonpositive %
Joint high	Cooling objective	CQR	4.237	12.222	4.939	0.000
Joint high	Cooling objective	ET-bag	5.316	15.640	6.132	0.000
Joint high	Cooling objective	ET-seed	5.269	15.322	6.009	0.000
Joint high	Cooling objective	GP-ARD	2.837	8.493	3.388	0.000
Joint high	Cooling objective	Log-poly	3.040	8.804	3.711	0.000
Joint high	Cooling objective	XGB	4.800	12.580	5.179	0.000
Joint high	Pressure drop	CQR	8.518	26.025	10.615	0.000
Joint high	Pressure drop	ET-bag	10.895	35.938	13.150	0.000
Joint high	Pressure drop	ET-seed	10.122	35.574	12.535	0.000
Joint high	Pressure drop	GP-ARD	8.190	25.206	9.927	0.000
Joint high	Pressure drop	Log-poly	14.018	46.858	16.127	0.000
Joint high	Pressure drop	XGB	8.243	26.024	10.408	0.000
Joint low	Cooling objective	CQR	4.270	13.776	5.013	0.000
Joint low	Cooling objective	ET-bag	4.626	15.417	5.553	0.000
Joint low	Cooling objective	ET-seed	4.680	15.718	5.638	0.000
Joint low	Cooling objective	GP-ARD	3.011	9.109	3.638	0.000
Joint low	Cooling objective	Log-poly	3.316	9.974	3.976	0.000
Joint low	Cooling objective	XGB	4.039	13.188	4.890	0.000
Joint low	Pressure drop	CQR	24.052	47.417	29.420	0.000
Joint low	Pressure drop	ET-bag	28.358	51.509	35.095	0.000
Joint low	Pressure drop	ET-seed	28.109	51.838	34.837	0.000
Joint low	Pressure drop	GP-ARD	11.046	30.784	13.720	0.000
Joint low	Pressure drop	Log-poly	13.950	36.317	16.678	0.000
Joint low	Pressure drop	XGB	23.198	44.652	27.442	0.000
Design random	Cooling objective	CQR	2.666	9.090	3.454	0.000
Design random	Cooling objective	ET-bag	3.952	12.804	4.893	0.000
Design random	Cooling objective	ET-seed	3.804	12.436	4.788	0.000
Design random	Cooling objective	GP-ARD	1.883	6.019	2.321	0.000
Design random	Cooling objective	Log-poly	2.042	6.567	2.550	0.000
Design random	Cooling objective	XGB	2.724	9.018	3.452	0.000
Design random	Pressure drop	CQR	8.186	30.602	11.317	0.000
Design random	Pressure drop	ET-bag	11.659	37.854	14.688	0.000
Design random	Pressure drop	ET-seed	10.938	36.264	14.020	0.000
Design random	Pressure drop	GP-ARD	5.908	20.230	7.608	0.000
Design random	Pressure drop	Log-poly	8.191	27.515	10.431	0.000
Design random	Pressure drop	XGB	8.023	28.270	10.543	0.000
x0 high	Cooling objective	CQR	2.590	8.963	3.300	0.000
x0 high	Cooling objective	ET-bag	3.443	11.493	4.357	0.000
x0 high	Cooling objective	ET-seed	3.371	11.166	4.259	0.000

Fold	Target	Model	Median APE %	95th APE %	Mean sMAPE %	Nonpositive %
x0 high	Cooling objective	GP-ARD	2.245	7.081	2.788	0.000
x0 high	Cooling objective	Log-poly	2.461	7.768	3.072	0.000
x0 high	Cooling objective	XGB	2.530	8.576	3.214	0.000
x0 high	Pressure drop	CQR	7.197	25.844	9.792	0.000
x0 high	Pressure drop	ET-bag	8.848	33.720	12.163	0.000
x0 high	Pressure drop	ET-seed	8.376	32.578	11.545	0.000
x0 high	Pressure drop	GP-ARD	7.288	24.285	9.071	0.000
x0 high	Pressure drop	Log-poly	11.149	38.899	13.437	0.000
x0 high	Pressure drop	XGB	6.952	25.107	9.190	0.000
x0 low	Cooling objective	CQR	4.992	16.452	6.010	0.000
x0 low	Cooling objective	ET-bag	6.154	20.781	7.341	0.000
x0 low	Cooling objective	ET-seed	6.174	20.603	7.311	0.000
x0 low	Cooling objective	GP-ARD	2.726	8.864	3.402	0.000
x0 low	Cooling objective	Log-poly	2.808	9.200	3.497	0.000
x0 low	Cooling objective	XGB	5.029	16.306	5.950	0.000
x0 low	Pressure drop	CQR	16.870	46.095	22.585	0.000
x0 low	Pressure drop	ET-bag	19.850	50.450	26.134	0.000
x0 low	Pressure drop	ET-seed	19.433	49.873	25.526	0.000
x0 low	Pressure drop	GP-ARD	8.477	27.956	10.763	0.000
x0 low	Pressure drop	Log-poly	12.050	33.358	14.736	0.000
x0 low	Pressure drop	XGB	15.960	41.844	20.815	0.000
x0 middle	Cooling objective	CQR	3.014	9.023	3.627	0.000
x0 middle	Cooling objective	ET-bag	4.490	11.669	5.174	0.000
x0 middle	Cooling objective	ET-seed	4.352	11.591	5.058	0.000
x0 middle	Cooling objective	GP-ARD	1.854	5.871	2.283	0.000
x0 middle	Cooling objective	Log-poly	2.011	6.286	2.480	0.000
x0 middle	Cooling objective	XGB	3.045	8.813	3.659	0.000
x0 middle	Pressure drop	CQR	8.674	29.656	11.405	0.000
x0 middle	Pressure drop	ET-bag	13.817	39.327	15.979	0.000
x0 middle	Pressure drop	ET-seed	12.901	37.723	15.183	0.000
x0 middle	Pressure drop	GP-ARD	6.026	19.590	7.607	0.000
x0 middle	Pressure drop	Log-poly	7.806	27.212	10.104	0.000
x0 middle	Pressure drop	XGB	8.862	28.025	11.108	0.000
x12 high	Cooling objective	CQR	2.879	9.796	3.711	0.000
x12 high	Cooling objective	ET-bag	3.848	13.754	4.957	0.000
x12 high	Cooling objective	ET-seed	3.699	13.340	4.826	0.000
x12 high	Cooling objective	GP-ARD	1.964	6.375	2.494	0.000
x12 high	Cooling objective	Log-poly	2.217	6.790	2.739	0.000
x12 high	Cooling objective	XGB	2.790	9.537	3.629	0.000
x12 high	Pressure drop	CQR	7.943	29.251	11.226	0.000
x12 high	Pressure drop	ET-bag	10.422	36.592	14.102	0.000
x12 high	Pressure drop	ET-seed	9.856	35.074	13.546	0.000
x12 high	Pressure drop	GP-ARD	5.948	19.623	7.637	0.000
x12 high	Pressure drop	Log-poly	9.106	30.176	10.987	0.000
x12 high	Pressure drop	XGB	7.784	27.449	10.587	0.000
x12 low	Cooling objective	CQR	2.830	9.581	3.575	0.000

Fold	Target	Model	Median APE %	95th APE %	Mean sMAPE %	Nonpositive %
x12 low	Cooling objective	ET-bag	4.039	12.629	4.993	0.000
x12 low	Cooling objective	ET-seed	3.920	12.341	4.859	0.000
x12 low	Cooling objective	GP-ARD	2.025	6.517	2.531	0.000
x12 low	Cooling objective	Log-poly	2.215	6.803	2.689	0.000
x12 low	Cooling objective	XGB	2.816	9.516	3.594	0.000
x12 low	Pressure drop	CQR	8.448	31.192	11.684	0.000
x12 low	Pressure drop	ET-bag	12.109	39.043	15.149	0.000
x12 low	Pressure drop	ET-seed	11.281	37.471	14.317	0.000
x12 low	Pressure drop	GP-ARD	6.230	20.156	7.827	0.000
x12 low	Pressure drop	Log-poly	8.185	27.347	10.441	0.000
x12 low	Pressure drop	XGB	8.669	29.458	11.043	0.000
x12 middle	Cooling objective	CQR	2.733	9.277	3.517	0.000
x12 middle	Cooling objective	ET-bag	4.003	12.349	4.900	0.000
x12 middle	Cooling objective	ET-seed	3.838	12.101	4.754	0.000
x12 middle	Cooling objective	GP-ARD	1.845	5.994	2.316	0.000
x12 middle	Cooling objective	Log-poly	1.975	6.446	2.490	0.000
x12 middle	Cooling objective	XGB	2.776	9.197	3.510	0.000
x12 middle	Pressure drop	CQR	8.083	29.937	11.392	0.000
x12 middle	Pressure drop	ET-bag	11.690	39.287	15.002	0.000
x12 middle	Pressure drop	ET-seed	11.217	37.510	14.278	0.000
x12 middle	Pressure drop	GP-ARD	6.001	20.362	7.639	0.000
x12 middle	Pressure drop	Log-poly	8.057	27.651	10.435	0.000
x12 middle	Pressure drop	XGB	8.221	28.671	10.766	0.000
x6 high	Cooling objective	CQR	3.467	11.956	4.377	0.000
x6 high	Cooling objective	ET-bag	4.564	16.242	5.772	0.000
x6 high	Cooling objective	ET-seed	4.391	15.422	5.541	0.000
x6 high	Cooling objective	GP-ARD	2.107	6.947	2.668	0.000
x6 high	Cooling objective	Log-poly	2.216	7.135	2.764	0.000
x6 high	Cooling objective	XGB	3.507	11.552	4.338	0.000
x6 high	Pressure drop	CQR	9.227	33.071	12.291	0.000
x6 high	Pressure drop	ET-bag	13.959	42.132	16.512	0.000
x6 high	Pressure drop	ET-seed	12.935	40.593	15.714	0.000
x6 high	Pressure drop	GP-ARD	6.020	20.816	7.756	0.000
x6 high	Pressure drop	Log-poly	8.743	30.123	11.065	0.000
x6 high	Pressure drop	XGB	9.227	31.178	11.757	0.000
x6 low	Cooling objective	CQR	4.216	10.729	4.839	0.000
x6 low	Cooling objective	ET-bag	6.015	13.953	6.732	0.000
x6 low	Cooling objective	ET-seed	5.720	13.581	6.399	0.000
x6 low	Cooling objective	GP-ARD	2.043	6.297	2.489	0.000
x6 low	Cooling objective	Log-poly	2.063	6.501	2.541	0.000
x6 low	Cooling objective	XGB	4.210	10.287	4.754	0.000
x6 low	Pressure drop	CQR	8.293	30.996	11.594	0.000
x6 low	Pressure drop	ET-bag	9.894	36.093	14.000	0.000
x6 low	Pressure drop	ET-seed	9.615	34.846	13.491	0.000
x6 low	Pressure drop	GP-ARD	6.250	20.873	8.075	0.000
x6 low	Pressure drop	Log-poly	8.257	26.736	10.467	0.000

Fold	Target	Model	Median APE %	95th APE %	Mean sMAPE %	Nonpositive %
x6 low	Pressure drop	XGB	7.797	28.003	10.655	0.000
x6 middle	Cooling objective	CQR	2.835	9.254	3.638	0.000
x6 middle	Cooling objective	ET-bag	3.824	12.798	4.879	0.000
x6 middle	Cooling objective	ET-seed	3.778	12.227	4.763	0.000
x6 middle	Cooling objective	GP-ARD	1.846	6.072	2.335	0.000
x6 middle	Cooling objective	Log-poly	2.097	6.718	2.604	0.000
x6 middle	Cooling objective	XGB	2.965	9.664	3.750	0.000
x6 middle	Pressure drop	CQR	7.933	29.491	11.207	0.000
x6 middle	Pressure drop	ET-bag	11.397	36.999	14.526	0.000
x6 middle	Pressure drop	ET-seed	10.665	34.925	13.716	0.000
x6 middle	Pressure drop	GP-ARD	5.905	19.359	7.458	0.000
x6 middle	Pressure drop	Log-poly	8.281	27.503	10.352	0.000
x6 middle	Pressure drop	XGB	7.767	28.218	10.496	0.000
Supplied random	Cooling objective	CQR	2.757	8.254	3.309	0.000
Supplied random	Cooling objective	ET-bag	3.913	12.460	4.704	0.000
Supplied random	Cooling objective	ET-seed	3.858	12.240	4.572	0.000
Supplied random	Cooling objective	GP-ARD	1.813	5.844	2.284	0.000
Supplied random	Cooling objective	Log-poly	2.014	6.520	2.491	0.000
Supplied random	Cooling objective	XGB	2.691	8.056	3.246	0.000
Supplied random	Pressure drop	CQR	7.940	27.866	10.769	0.000
Supplied random	Pressure drop	ET-bag	11.048	35.691	14.132	0.000
Supplied random	Pressure drop	ET-seed	10.547	33.988	13.432	0.000
Supplied random	Pressure drop	GP-ARD	5.790	19.528	7.399	0.000
Supplied random	Pressure drop	Log-poly	7.663	26.213	9.904	0.000
Supplied random	Pressure drop	XGB	7.417	25.541	9.689	0.000